

Sld3CBD-Cdc45 structural insights into Cdc45 recruitment for CMG complex formation

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eLife Assessment

This **valuable** paper describes the crystal structure of a complex of Sld3-Cdc45-binding domain (CBD) with Cdc45, which is essential for the assembly of an active Cdc45-MCM-GINS (CMG) double hexamers at the replication origin. Although the results shown in the paper are of interest to researchers in DNA replication and genome stability, the biochemical analysis of protein-protein interaction and DNA binding is **incomplete**, and the paper needs additional data.

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Abstract

DNA replication requires Cdc45 and GINS recruitment into the MCM duplex hexamer by initiation factors to form an active helicase, the Cdc45-MCM-GINS (CMG) complex, at the replication origins. The initiation factor Sld3 is a central regulator of Cdc45 and GINS recruitment worked with Sld7 together. However, the mechanism through which Sld3 regulates CMG complex formation remains unclear. Here, we present the structure of the Sld3 Cdc45-binding-domain in complex with Cdc45 (Sld3CBD-Cdc45), showing detailed interactions and conformational changes required for binding to each other. The mutant analysis indicated that the binding between Sld3CBD and Cdc45 could be broken easily. We also revealed that Sld3CBD, GINS, and MCM bind to different sites on Cdc45 in the Sld3CBD-CMG model, indicating that after recruitment of Cdc45, Sld7-Sld3 could remain in Cdc45-MCM until CMG formation. The consistency between the particle size of Sld7-Sld3-Cdc45 and the distance between Sld3CBDs in the Cdc45-MCM dimer indicated the binding manner of the Cdc45-Sld3-[Sld7]₂-Sld3-Cdc45 off/on MCM duplex hexamer. A DNA-binding assay of Sld3 and its complexes with single-stranded ARS1 fragments revealed a relationship between the dissociation of Sld7-Sld3 from CMG and the unwound single-stranded DNA. These findings help to further our understanding of the molecular basis of regulation of CMG complex formation by Sld3.

Introduction

Eukaryotic chromosomal DNA replication in the cell cycle, a tightly regulated process that ensures precise gene copying, begins with the unwinding of double-stranded DNA (dsDNA) into single-stranded DNA (ssDNA), including the formation of an active helicase Cdc45–MCM–GINS (CMG) complex [1, 2]. In the G1 phase of the yeast cell cycle, two inactive helicase Mcm2–7 hexamer rings (Mcm2–6–4–7–3–5) are loaded onto autonomously replicating sequences (ARSs) of dsDNA to form a double hexamer (MCM DH) at the replication origin [3–5]. To facilitate this loading, Cdt1 binds to the Mcm2–7 hexamer ring and stabilizes a gap formation between the Mcm2 and Mcm5, which serves as a dsDNA entry gate [6–9]. Next, the MCM DH on dsDNA is phosphorylated by Dbf4-dependent protein kinase (Cdc7 kinase or DDK) (Supplementary Figure 1A), allowing the key factor Sld3 and its partner Sld7 to recruit Cdc45 (Sld7–Sld3–Cdc45) to each MCM (Supplementary Figure 1B) [10–13]. Subsequently, the phosphorylation of Sld3 by cyclin-dependent kinase (CDK) regulates the assembly of GINS onto MCM DH with Dpb11, CDK-phosphorylated Sld2, and DNA polymerase ϵ to form CMG (Supplementary Figure 1C) [14–19]. Finally, the factors Sld2, Sld3, Sld7, and Dpb11 are released by elusive mechanisms from an active CMG and bidirectional replication by translocating along the 3' to 5' direction of the DNA strand (Supplementary Figure 1D) is facilitated [20, 21].

As a central regulator of CMG formation in *Saccharomyces cerevisiae*, Sld3 functions as a bridge protein in the Sld7–Sld3–Cdc45 complex to recruit Cdc45 to DDK-phosphorylated MCM DH. This Sld3 contains three domains each of which binds mainly to one of Sld7, Cdc45, and MCM. The N-terminal domain of Sld3 (Sld3NTD: M1–L116) binds to the N-terminal domain (M1–D130) of Sld7 (Sld7NTD) (Sld7 C-terminal domain [Sld7CTD: K168–S257] is a self-dimerization domain) [22]. Following the NTD of Sld3, the middle part is a central portion known as the Cdc45-binding domain (Sld3CBD: S148–K430) [12, 23]. A previous study also demonstrated that a region (L510–R530) in the Sld3 C-terminal domain (Sld3CTD: T445–T679) binds to Mcm4 and 6 [24]. CDK-dependent phosphorylation of two residues (T600 and S622) downstream of Sld3CTD initiates the binding of GINS-carrying Dpb11 to Cdc45–MCM [19, 25]. In addition to its role as a bridge in the recruitment of Cdc45 and GINS, Sld3 binds to two single-stranded DNA (ssDNA) fragments of ARS1 which was identified as an origin of DNA replication (ssARS1-2 and ssARS1-5), but not to the corresponding double-stranded ARS1 (dsDNA) [26]. The specific association between Sld3 and ssDNA is not affected by CDK phosphorylation. Furthermore, the structures of the Sld7CTD dimer (PDBID:3X38) [22], Sld7NTD–Sld3NTD (PDBID:3X37) [22], Sld3CBD (PDBID:3WI3) [23], MCM DH (6F0L) [27], CMG (PDBIDs:3JC5, 3JC6, 3JC7) [28], CMG–DNA–polI ϵ (PDBID:7Z13) [29], CMG–DONSON–DNA (PDBID:8Q60) [30], and so on, were determined through crystallography and cryogenic electron microscopy (cryo-EM). Cdc45 belongs to the DHH superfamily of proteins defined by the conserved triad motif DHH (Asp–His–His), and there is a DHH-associate domain (DHHA1: R523–L650) at the C-terminus [31]. Recent single-molecule biochemical assays have reported the stepwise recruitment of multiple Cdc45s to the MCM DH [32]. However, how Sld3–Sld7 brings Cdc45 into the MCM for the formation of CMG to regulate the initiation of DNA replication remains unclear.

In the present study, we aimed to understand how Sld3 recruits Cdc45 to the MCM DH with Sld7 for CMG formation through structure and particle analyses. We determined the structure of *S. cerevisiae* Sld3CBD–Cdc45 at 2.6 Å resolution and presented the detailed interactions between Sld3 and Cdc45, confirmed through *in vitro* and *in vivo* mutant analyses. Compared to the monomer structures, the conformation of Sld3CBD and Cdc45 in the Sld3CBD–Cdc45 complex changed significantly to bind each other. Based on the structural similarity of Cdc45 in Sld3CBD–Cdc45 and CMG, we modelled Sld3CBD–Cdc45–MCM–dsDNA and SCMG–dsDNA (Sld3CBD–CMG–dsDNA) as a snapshot of how helicase CMG forms. The models demonstrated that Sld3CBD, MCM, and GINS bind to different sites on Cdc45, indicating that Sld7–Sld3 could remain at Cdc45–MCM until CMG

formation after GINS loading. Consistency between the particle size of Sld7–Sld3ΔC–Cdc45, (Sld3ΔC: M1–K430; truncated the C-terminal domain) as per spectroscopic analysis, and the distance of Sld3CBDs in the Cdc45–MCM dimer suggested that the ternary complex of Sld7–Sld3–Cdc45 forms a dimer off/on the MCM DH for recruiting Cdc45. Furthermore, ssDNA-binding analysis of ARS1 fragments implied that the release of Sld3–Sld7 could be associated with ssARS1 unwound by CMG. Our findings illustrate the recruit–release function of Sld3 in CMG formation and expand our knowledge of the initiation process of DNA replication.

Results

The overall structure of Sld3CBD and Cdc45 complex

We obtained a recombinant Sld3CBD–Cdc45 complex of *S. cerevisiae* and determined its crystal structure at 2.6 Å resolution using molecular replacement (**Figure 1A** [\[23\]](#), Supplementary Figure 2 A, 3). Only one complex exists within an asymmetric unit. Similar to monomer Sld3CBD (PDBID: 3WI3 with RMSD of 0.50 Å for 181 Ca atoms) [\[23\]](#), Sld3CBD (Y154–P420) in the Sld3CBD–Cdc45 complex is an α-helical structure with two disordered regions of R317–S336 and P364–A369, but with a significant difference in that a disordered part in monomer Sld3CBD was visualized as a C-terminal part of bent long helix α8 (F294–R316; referred to as α8CTP) (Supplementary Figures 3, 4A and 5). Cdc45 (M1–L650) is an α/β structure composed of three β-sheets (anti-parallel: β1–β6–β5–β4–β2–β3, anti-parallel: β7–β8, mixed: β9–β10–β11–β13–β12) surrounded by 21 α-helices (Supplementary Figure 3). Owing to the poor electron density map, the regions D106–K110, D166–K227, S306–V310, and T438–D460 on the molecular surface could not be built. In contrast to the monomer human Cdc45 (huCdc45) and the CMG form (Cdc45 in the yeast CMG complex), more than the N-terminal half of the protruding long helix α7 was disordered in the Sld3CBD–Cdc45 complex (Supplementary Figures 4B and 6).

Conformation changes in Sld3CBD and Cdc45 for binding each other

Sld3CBD binds to Cdc45 in a way similar to a toothed gear (**Figure 1B** [\[23\]](#)), with a contact surface of 1808 Å² accounting for approximately 13.3% and 7.0% of the total surface of Sld3CBD and Cdc45, respectively. Two helices (α8 and α9) of Sld3CBD formed a plier structure and gripped the C-terminal domain DHHA1 (R523–L650) of Cdc45 through hydrophobic interactions and hydrogen bonds. The C-terminal loop (L646–L650) and the C-terminal part of α7 (E232–S242) of Cdc45 sandwiched the helix α8CTP of Sld3CBD. Compared to the isolated form (PDBIDs: 5DGO for huCdc45 [\[31\]](#) [\[33\]](#) and 6CC2 for EhCdc45 [\[33\]](#)) and the CMG form (PDBID: 3JC6 [\[28\]](#)), the Cdc45 in the Sld3CBD–Cdc45 complex changed the conformation of DHHA1 to form pockets on its two sides for binding Sld3CBD α8CTP and α9. The remaining part of Cdc45 (~K517) retained a structure with RMSD values of 1.29 Å (isolated huCdc45), 1.81 Å (isolated EhCdc45), and 1.295 Å (in CMG) for 243, 251, and 361 Ca atoms, respectively.

The helix α8CTP (F294–R316) of Sld3CBD was surrounded by the C-terminal domain DHHA1 and a C-terminal part of the protruded long helix α7 (E232–S242) of Cdc45 but completely disordered in Sld3CBD alone (PDBID: 3WI3) [\[23\]](#) (**Figure 1B** [\[23\]](#) and Supplementary Figure 4A). Therefore, the helix α8CTP seems to be an intrinsically disordered segment when Sld3 alone but folds into a helix coupled to the binding partner Cdc45 in the Sld3CBD–Cdc45 complex. This α8CTP is essential for binding to Cdc45, as its deletion inhibited cell growth in a previous study [\[23\]](#). Furthermore, proline substitution for Cdc45 Ser242 (Cdc45-35), which interacts with L307 and T310 in Sld3CBD α8CTP (**Figure 1C** [\[23\]](#)), conferred temperature-sensitive growth to yeast cells (Supplementary Figure 7). Although an increase in the number of copies of Sld3–Sld7 could weakly suppress cell growth defects, it did not recover the disrupted interaction.

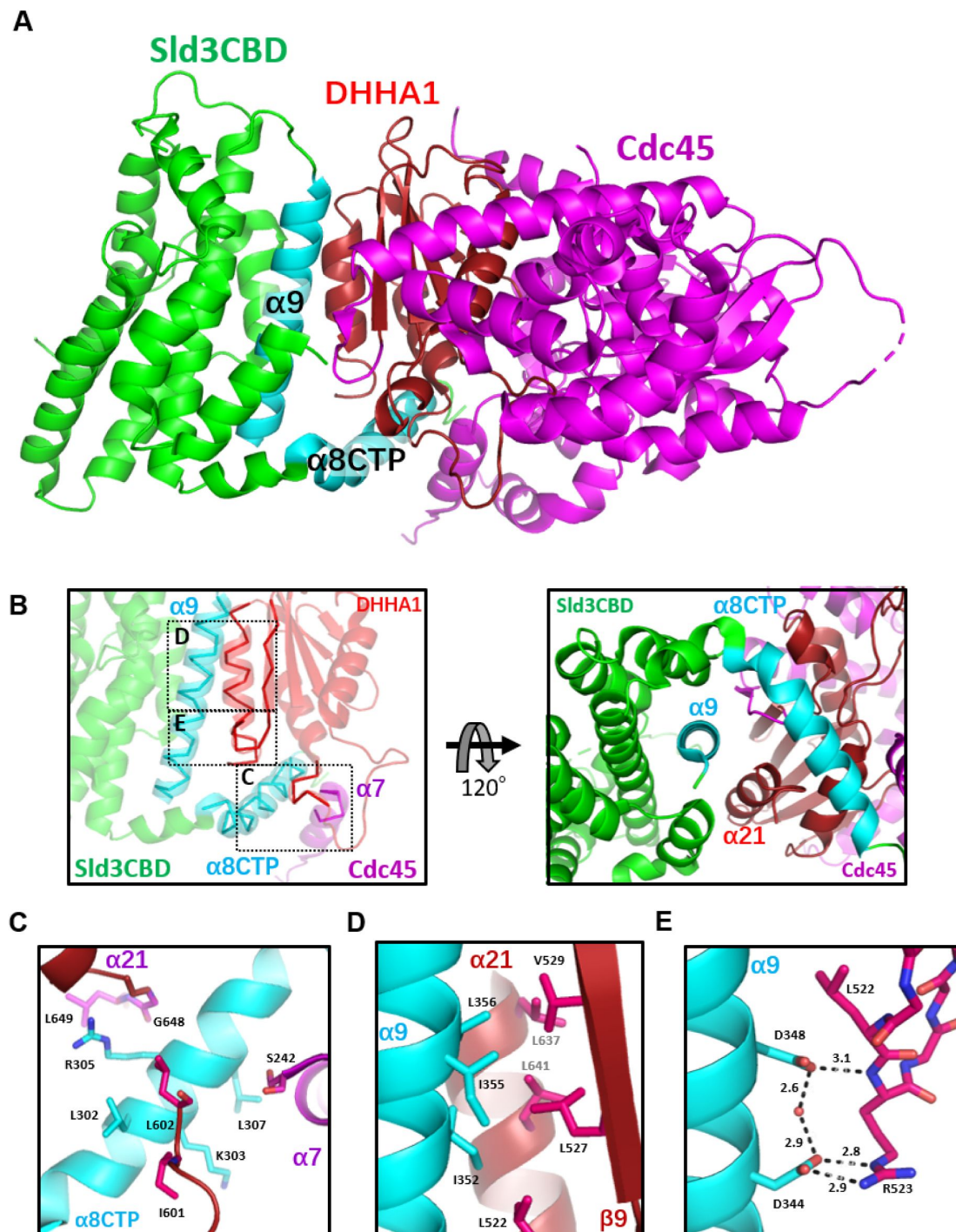


Figure 1.

Sld3CBD-Cdc45 complex structure

(A) Structure of the Sld3CBD-Cdc45 complex. Sld3CBD and Cdc45 are colored in green and magenta, respectively. Cdc45-binding parts α 8CTP and α 9 are labelled and colored in cyan. The DHHA1 of Cdc45 is labelled and colored in dark magenta. (B) Binding part of Sld3CBD-Cdc45 in different viewing. Dotted squares C, D, and E mark three binding sites, corresponding to the bottom panel. (C) Binding site on the α 8CTP of Sld3CBD. (C) (D) Two binding sites involving hydrophobic and hydrogen-bond interactions on the α 9 of Sld3CBD, respectively. The interacted residues are depicted by sticks and labelled. The black dotted lines show hydrogen bonds.

The $\alpha 9$ (L337–E360) of Sld3CBD is the secondary Cdc45-binding region, which is located in a shallow dent formed by the C-terminal helix and a loop with the following β -strand (K520–Q531) of Cdc45 DHHA1 (**Figure 1B**, **1D** and **1E**). Three hydrophobic residues (I352, I355, and L356) in Sld3CBD $\alpha 9$ interact with the C-terminal sheet (L522, L527 and V529) and helix (L641, and L647) of Cdc45 DHHA1 (**Figure 1D**), while the side chains of two negatively charged residues (D344 and D348) form hydrogen bonds through a water molecule to the main and side chains of R523 on a loop of Cdc45 DHHA1 (**Figure 1E**). A disordered region, L527–V529, of Cdc45 DHHA1 in the isolated form forms a β -sheet in the Sld3CBD–Cdc45 complex and binds to the I352 and I355 of Sld3CBD $\alpha 9$. We substituted single, double, or ternary positively charged or hydrophilic residues at D344, D348, I352, I355, and L356 of Sld3CBD $\alpha 9$ (D344R/D348R: Sld3-2R, I352Y: Sld3-Y, I352S/I355S/L356S: Sld3-3S, I352E/I355E/L356E: Sld3-3E). Based on the CD spectra, we confirmed that these mutants retained the structural elements of Sld3CBD (Supplementary Figure 8). Double and ternary mutants Sld3-2R, Sld3-3S, and Sld3-3E eliminated Cdc45-binding affinity, whereas the single mutant Sld3-Y seemed to retain a faint interaction with Cdc45 (**Figure 2A**). We also substituted the single, double, or ternary residues R523, L522, L527, V529, L637, and L641 of Cdc45 on Sld3CBD $\alpha 9$ binding sites (L637S/L641S: Cdc45-IIS, L637E/L641E: Cdc45-IIE, L522S/L527S/V529S: Cdc45-IIIS, L522E/L527E/V529E: Cdc45-IIIE, and Cdc45-A: R523A). All Cdc45 mutants disrupted the binding between Sld3CBD and Cdc45, except for Cdc45-A, as surmised from the Sld3CBD–Cdc45 structure (**Figure 2B**). Although mutant Cdc45-A has reduced three hydrogen bonds with D344 of Sld3CBD, the remaining hydrogen-bond network maintains contact between Sld3CBD and Cdc45. Furthermore, *in vivo* genetic studies confirmed the importance of these Sld3 residues. By expressing Sld3-3S, Sld3-3E, and Sld3-2R in Sld3, the variant strains caused complete no growth, whereas the Sld3-Y strain maintains cell growth (**Figure 2C**), indicating that these residues are required to work together to bind Cdc45 and the loss of the Cdc45-binding affinity of Sld3 inhibits the recruitment of Cdc45 and the subsequent formation of the active replicative helicase CMG for DNA replication.

In comparison with Cdc45 alone (huCdc45) or CMG-form (in CMG complex), domain DHHA1 of Cdc45 changed conformation significantly for binding to Sld3CBD (Supplementary Figure 4C). The loop I595–N604 in Cdc45 DHHA1 changed conformation to interact with Sld3CBD $\alpha 8$ CTP. Subsequent the following helix $\alpha 19$ (F605–E615) rotated the C-terminus by 25 degrees, which altered the conformation of the next two β -strands ($\beta 12$ and $\beta 13$) in the mixed β -sheet ($\beta 9$ – $\beta 10$ – $\beta 11$ – $\beta 13$ – $\beta 12$) (Supplementary Figure 3 and 4C), allowing Sld3CBD $\alpha 8$ CTP to enter the binding pocket. Interestingly, the Sld3CBD–Cdc45 structure showed that the Sld3CBD binding site of Cdc45 completely differed from that of GINS and MCM bound to Cdc45, suggesting that the Sld3CBD, Cdc45 and GINS could bind to MCM together (Supplementary Figure 9A). Furthermore, we conducted a mutation analysis on two Cdc45 residues involved in binding to MCM (Cdc45 G367D) and GINS (Cdc45 W481R) [34], and found that these mutations did not disrupt the binding of Sld3CBD–Cdc45 complex (Supplementary Figure 9B).

Cdc45 recruitment to MCM DH by Sld3 with partner Sld7

Except for the Sld3 binding region DHHA1, the N-terminal domain of Cdc45 (Cdc45NTD) maintained a structure similar to that of the monomer, Sld3CBD–Cdc45 complex, and CMG complex. Therefore, we modelled Sld3CBD–Cdc45–MCM–dsDNA and SCMG–dsDNA (**Figure 3**) by superposing Cdc45NTD structures between Sld3CBD–Cdc45 and CMG dimers respectively, which were modelled by superposing Mcm2–7 structures between CMG (PDBID: 3JC6) [28] and MCM–dsDNA (PDBID: 5BK4) [35]. In the models, two Sld3CBDs are located in each monomer of the Cdc45–MCM dimer over 230 Å apart (**Figure 3A**).

To investigate how Sld7–Sld3 brings Cdc45 to MCM DH, we attempted particle analysis for the Sld7–Sld3–Cdc45 complex using the purified recombinant Sld7–Sld3 Δ C–Cdc45. The approximate molecular weight of Sld7–Sld3 Δ C–Cdc45 was estimated to be >400 kDa according to a weight calibration of size-exclusion chromatography (Supplementary Figure 2B). Given that the molecular weight calculated from their amino acid sequences was 158 kDa, the purified complex should be a

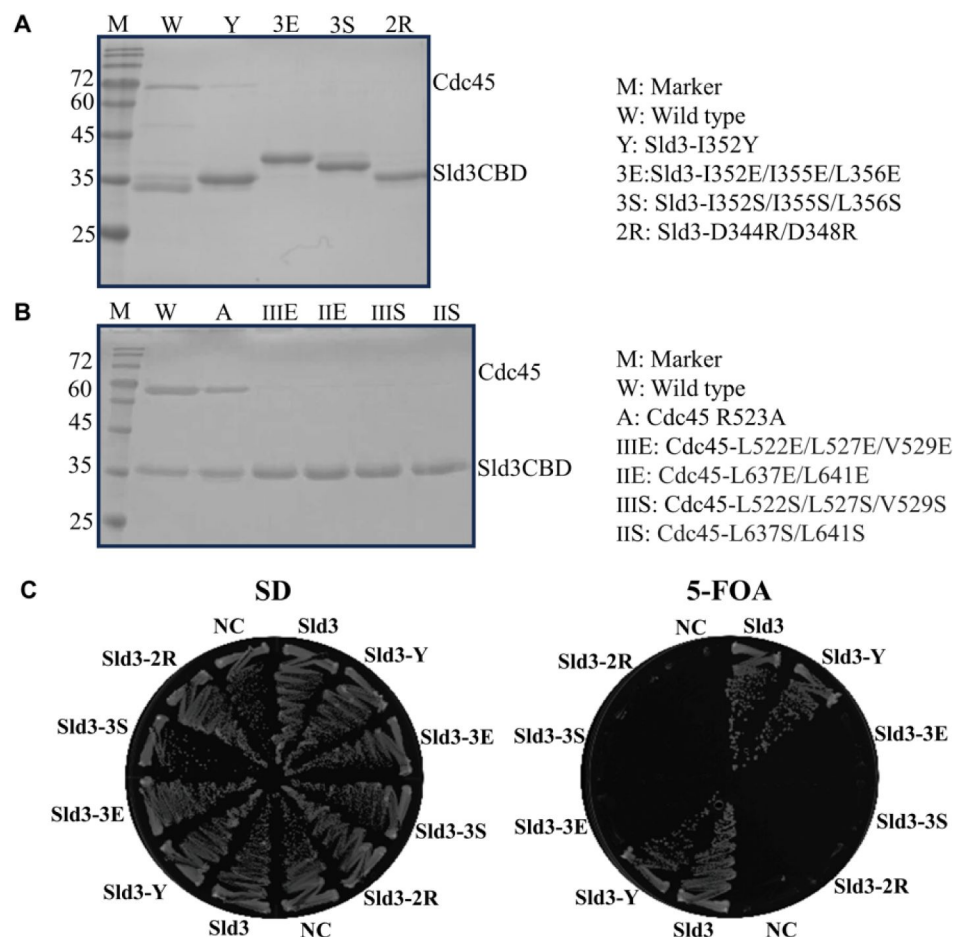


Figure 2.

Mutation analysis of interacted residues

(A) *In vitro* binding analysis was checked using SDS-PAGE after Ni-affinity chromatography extraction of co-overexpressed Cdc45 with each Sld3 mutant. Sld3 was tagged by His-tag to bind to the column. The labels M, W, Y, 3E, 3S, and 2R are explained on the right. (B) *In vitro* binding analysis was checked using SDS-PAGE after Ni-affinity chromatography extraction of co-overexpressed Sld3 with each Cdc45 mutant. Sld3 was tagged by His-tag to bind to the column. The labels M, W, A, III E, IIE, IIIS, and IIS are explained on the right. (C) *In vivo* cell growth analysis of yeast cells carrying *SLD3* or its mutant plasmids. The yeast YYK13 cells carrying *SLD3* or its mutant plasmids were streaked onto SD and FOA plates and then incubated at 298 K for 3 days. YYK13 yeast is a mutant lacking the *SLD3* gene with added *SLD3/sld3* mutant gene (YCplac22 plasmid containing *SLD3* or *sld3* mutant) that grew on SD and FOA plates. The empty plasmids were used as negative control (NC). Mutations in Sld3-Y, Sld3-3E, Sld3-3S, and Sld3-2R are the same as those in A.

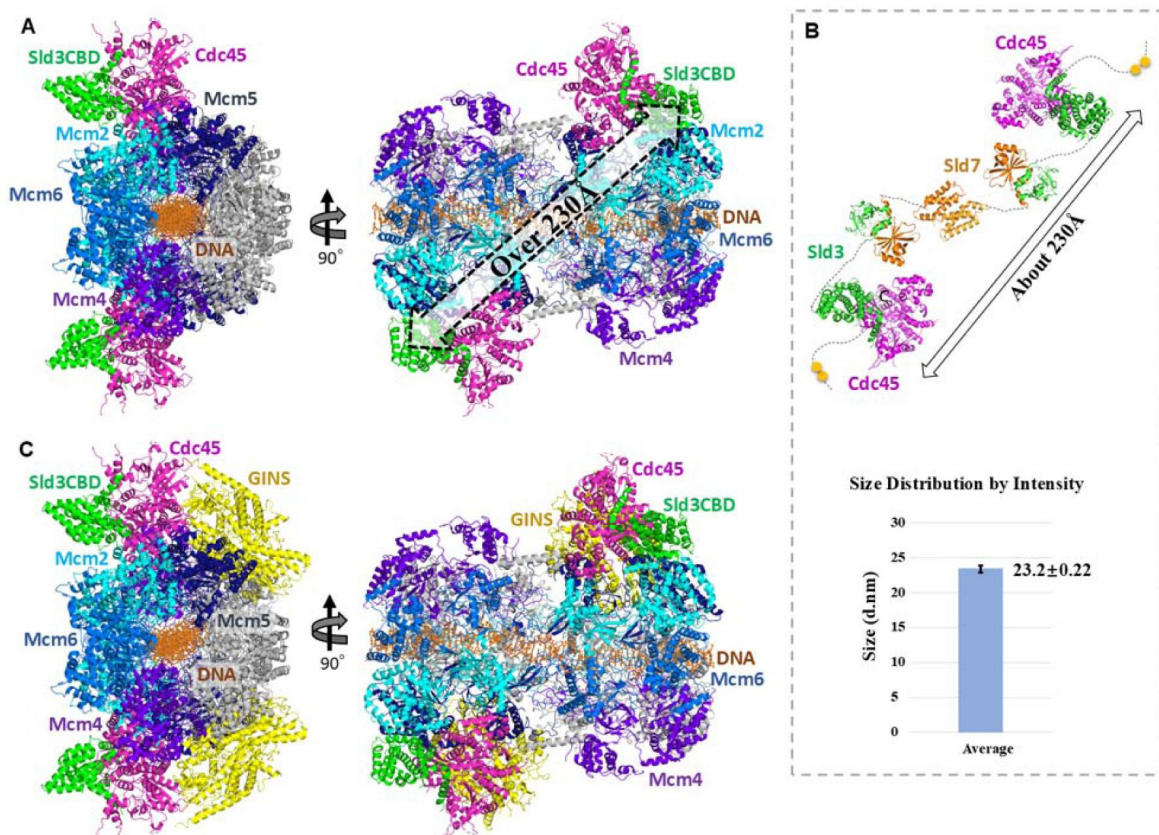


Figure 3.

Ribbon models of complexes in dimer form and particle analysis

(A) Sld3CBD-Cdc45-MCM-dsDNA complex. Mcm2, 5, 4, and 6 subunits are colored in cyan, blue, marine, and light blue, respectively. Subunits Mcm3 and Mcm7 are colored in gray. Green and pink are used to color Sld3CBD and Cdc45, respectively. dsDNA is represented by a dark-orange stick. (B) Sld3CBD-Sld7-Cdc45 dimer before associating with the MCM DH. Sld3 and Cdc45 are shown in the same color as they are in A, while Sld7 is colored in orange. The two phosphorylated Sld3 residues are depicted as yellow balls. Particle analysis of Sld7-Sld3ΔC-Cdc45 through dynamic light scattering is shown on the bottom panel. The average peak size of the particle size distribution of the Sld7-Sld3ΔC-Cdc45 complex was estimated to be 23.2 Å in diameter. The measurement was carried out independently four times (Supplementary Figure 9). (C) SCMG-dsDNA complex. GINS is shown in yellow and the remainder are colored identically to those in A and B.

dimer. Considering the Sld7–Sld3ΔC–Cdc45 dimer should be a long rod-shaped molecule, the estimated value from SEC could be larger than theoretical values. Subsequently, using dynamic light scattering, the particle size (hydrodynamic diameter) of the tripartite complex was estimated to be around 232 Å (**Figure 3B** [↗](#), Supplementary Figure 10), which is consistent with the distance of Sld3CBDs in the model of Sld3CBD–Cdc45–MCM dimer. Furthermore, considering that the domains of Sld7 (NTD: Sld3NTD-binding, CTD: self-dimerization) and Sld3 (NTD: Sld7NTD-binding, CBD: Cdc45-binding, CTD: MCM-binding) function independently [22 [↗](#), 23 [↗](#)], we estimated a dimer model of Sld7–Sld3–Cdc45, as shown in **Figure 3B** [↗](#). Because the domains of Sld7–Sld3–Cdc45 are linked by long loops, the dimer of Sld7–Sld3–Cdc45 is flexible with long-shape.

Our SCMG–dsDNA model demonstrated that Sld3CBD neighbors Mcm2 and binds to Cdc45 on the opposite side of GINS binding (**Figure 3C** [↗](#)), indicating that Sld7–Sld3 bound to the Cdc45–MCM dimer did not contact GINS and could be remained for CMG formation. A recent study also reported a structure of the DONSON (Sld2 homolog) dimer with CMG (8Q6O), showing that the DONSON dimer delivers GINS to MCM and reconfigures the MCM motors in the double CMG [30 [↗](#)]. The DONSON dimer is loading at the GINS site on MCM, which is the opposite of Sld3CBD in the SCMG–dsDNA model. Interestingly, in the Sld3CBD–Cdc45–MCM–dsDNA and SCMG–dsDNA model, we found that a part (S358–D383) of Sld3CBD containing the C-terminal of α9, a disordered fragment, and α10 was in contact distance with Mcm2CTD. In addition, the Sld3CBD-bound Cdc45 DHHA1 appeared to become close to Mcm2CTD. In contrast, the Cdc45 DHHA1 does not contact with Mcm2–7 or GINS in the CMG structure (Supplementary Figure 11). The conformational change in Cdc45 DHHA1 not only facilitates binding with Sld3CBD, but could also cause contact with Mcm2 without affecting the interaction between Cdc45 and GINS.

ssDNA binding affinity of Sld3 depended on complex formation with Cdc45 and Sld7

Previous studies showed that Sld3 binds directly to single-strand DNA fragments (ssARS1-2 and ssARS1-5) of ARS1 identified as an origin of DNA replication [26 [↗](#)]. ARS1 was divided into three 80-bp segments: ARS1-12, ARS1-34, and ARS1-56, and these dsDNA segments could unwind into six signal-stranded DNA fragments with 80b length: ssARS1-1, 2, 3, 4, 5, and 6 (Supplementary Figure 12A). Given that Sld3 binds to Sld7 and Cdc45 on the MCM–DNA complex during CMG formation, we investigated whether Sld7 and Cdc45 affect the ssDNA-binding affinity of Sld3. Therefore, we performed an ssDNA binding assay using the Sld3CBD, Sld3CBD–Cdc45, Sld7–Sld3ΔC–Cdc45 and Sld7–Sld3ΔC (Sld3ΔC: binding part M1–K430 to Sld7 and Cdc45) complexes against different regions of ssARS1 for comparison. Cdc45 binds tighter to long ssDNA (>60 bases) with a litter sequence specificity [36 [↗](#)]. Therefore, we fragmented each ssARS1 into fragments of 40 bases to prevent this nonspecific interaction (**Figure 4A** [↗](#) and Supplementary Figures 12B).

To investigate the specificity of the ssDNA binding affinity of Sld3, we employed an electrophoretic mobility shift assay (EMSA) using non-denaturing PAGE (native-PAGE) with ssDNAs or proteins alone as controls. For the Sld3CBD and ssDNA mixtures at a molar ratio of 1:1, the band corresponding to ssDNA disappeared, indicating a strong binding affinity (**Figure 4B** [↗](#)). The ssDNA band remained (**Figure 4B** [↗](#)) and new bands corresponding to the ssDNA–protein complex appeared in CBB staining PAGE (Supplementary Figures 13) when the Sld3CBD–Cdc45 complex was mixed with ssDNA at the same ratio, indicating that the binding affinity of Sld3CBD–Cdc45 for ssDNA was lower than that of Sld3CBD alone. Additionally, the disappearance of ssDNA bands and the appearance of new bands corresponding to the ssDNA–protein complex correlated with an increase in the concentration of Sld3CBD–Cdc45 in the mixture.

In the presence of the Sld7–Sld3ΔC–Cdc45 complex, the band corresponding to ssDNA disappeared in the equimolar ratio of protein and ssDNA, similar to that observed for Sld3CBD (**Figure 4B** [↗](#) and Supplementary Figures 13). This result demonstrates that the presence of Sld7 and Sld3NTD could increase the ssDNA-binding affinity to a level comparable to that of Sld3CBD. Also, Sld7–

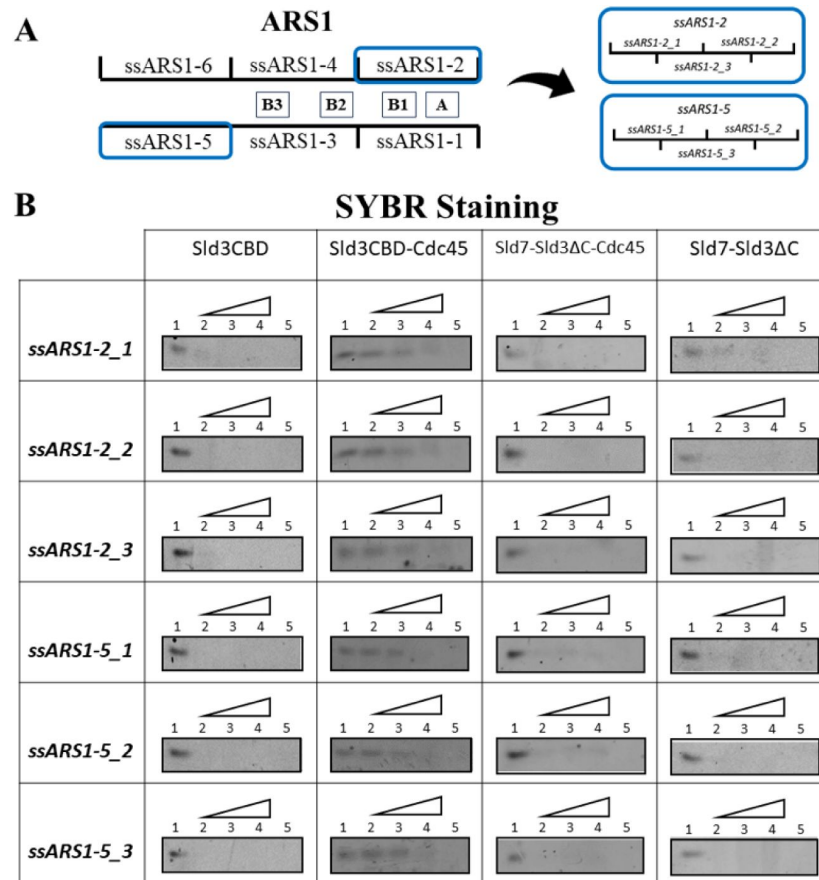


Figure 4.

Electrophoresis mobility shift assay (EMSA) of ssDNA binding to Sld3 and its complexes with Sld7 and Cdc45

(A) Schematic of ssARS1-1–ssARS1-6. ARS1 is identified as an origin of DNA replication, and divided into three 80-bp segments: ARS1-12, ARS1-34, and ARS1-56. These dsDNA segments could unwind into six signal-stranded DNA fragments with 80b length: ssARS1-1, 2, 3, 4, 5, and 6. The important elements of A (ARS consensus sequence), B1, B2, and B3 for unwinding are marked [37]. We divided ssARS1-2 and ssARS1-5 fragments (blue squares) into 40-base lengths for EMSA. (B) ssDNAs were visualized using SYBR safe on polyacrylamide gels. In the presence of ssDNA fragments, Sld3CBD, Sld3CBD–Cdc45, Sld7–Sld3ΔC–Cdc45 and Sld7–Sld3ΔC were incubated with molecular mass-related concentrations. The molecular ratio of ssDNA to protein in lanes 1, 2, 3, 4, and 5 was 1:0, 1:1, 1:2, 1:4, and 0:1, respectively. The controls for ssDNA and protein are lanes 1 and 5, respectively. The disappearance of the ssDNA band in lanes 2–5 indicates that the protein (Sld3CBD, Sld7–Sld3ΔC and Sld7–Sld3ΔC–Cdc45) binds with high affinity.

Sld3 Δ C showed a ssDNA-binding affinity similar to that of Sld7–Sld3 Δ C–Cdc45, implying that ssDNA-binding of Sld7–Sld3 Δ C is independent of Cdc45. Furthermore, the results revealed no significant difference in binding affinity between the 40-base fragments of ssARS1-2-1, 2, 3 and ssARS1-5-1, 2, 3 for Sld3CBD, Sld3CBD–Cdc45, Sld7–Sld3 Δ C–Cdc45, and Sld7–Sld3 Δ C, indicating that there is no stronger binding-region specific to ssARS1-2 or ssARS1-5 fragments. For sequence specificity, we also analyzed other fragments of ssARS1-1, 3, 4, and 6 and dsARS1, and all of them showed no binding (Supplementary Figure 14). Such binding experiments can serve as a negative control, confirming that the absence or presence of ssDNA bands is contingent on protein binding.

The surface charge of the Sld3CBD–Cdc45 structure, shows that Cdc45 covers the main positive charge region of Sld3CBD α 8CTP (Supplementary Figure 15A), which may weaken the binding affinity of Sld3CBD–Cdc45 to ssDNA. Conversely, on the Sld3–Sld7 structure, there is a large positive charge area strip on Sld7NTD (Supplementary Figure 15B). Considering that ssARS1 is unwound from dsARS1 by the activated helicase CMG complex formed after loading Cdc45 and GINS, Sld3–Sld7 having a stronger ssARS1-binding affinity may provide an advantage for the dissociation of Sld7–Sld3 from the CMG complex.

Discussion

As a central regulator of helicase CMG formation, Sld3 has attracted interest in studies aiming to understand the initiation of DNA replication. Owing to the lack of structural information on Sld3 complexed with Cdc45 or MCM, how Sld3 regulates helicase activation with other factors remains unknown. Here, we present the structure of the Sld3CBD–Cdc45 complex and particle size of the Sld7–Sld3–Cdc45 complex to examine the detailed mechanisms underlying CMG formation.

Sld3 is highly conserved in eukaryotes, whereas its functional homolog/counterpart in metazoans Treslin (also known as Ticrr) has a distinct size and sequence, except for Sld3CBD. Sequence alignment of Sld3CBD among Sld3 and Treslin revealed that all Cdc45-binding residues in α 8 and α 9 identified in our study were almost conserved or exhibited conserved changes (Supplementary Figures 4 and 5), indicating that these regions provide a similar mode of interaction between Sld3CBD and Cdc45 in the regulation of metazoan DNA replication. Therefore, we hypothesize that they load Cdc45, as observed in yeast Sld3 and Sld7.

Sld3CBD and Cdc45 significantly change their conformation to bind to each other in the Sld3CBD–Cdc45 complex compared with their conformation as monomers. Interestingly, the conformational changes in Cdc45 DHHA1 upon binding to Sld3CBD also caused contact with Mcm2NTD in the Sld3CBD–Cdc45–MCM–dsDNA and SCMG–dsDNA models, whereas DHHA1 interacted with neither MCM nor GINS in the CMG structure. Taken together, Sld3CBD contacts Mcm2NTD in the Sld3CBD–Cdc45–MCM–dsDNA and SCMG–dsDNA models and then seems to play a guiding role in helping Cdc45 bind to MCM at the right position. Sld3CBDs in each monomer of the Sld3CBD–Cdc45–MCM dimer were located at a distance of more than 230 Å, which is consistent with the results of a particle size analysis of the Sld7–Sld3 Δ C–Cdc45 complex off MCM DH in solution. Therefore, we propose that a binding manner of Sld7–Sld3–Cdc45 in a flexible long-sharped dimer Cdc45–Sld3–(Sld7)₂–Sld3–Cdc45 off/on MCM DH is advantageous for efficiently recruiting two Cdc45 molecules to an MCM DH, consequently leading to the formation of a pair of CMG helicases.

In the SCMG–dsDNA complex model, Cdc45 bound to Sld3CBD, MCM, and GINS on different sides (with contact surfaces of 6.7, 4.8, and 5.1% of the total Cdc45 surface, respectively). Our structure of Sld3CBD–Cdc45 and model shows that these bindings occur at different sites of Cdc45, suggesting that Sld3CBD, Cdc45 and GINS could bind to MCM together, and do not directly interfere with each other. The competition of Sld3 and GINS for binding to Cdc45 or Cdc45–MCM reported by using *in vitro* binding experiments [26], may be caused by the conformation change of Cdc45 DHHA1 or without other initiation factors. In particular, Sld3 and GINS bind to Cdc45 and MCM rings at

opposite positions (Mcm2–4–6 and Mcm5–3–7 vs. Sld3 and GINS, respectively), suggesting that the GINS-recruitment protein should cross a long-distance in an MCM monomer or MCM DH to recognize the phosphorylation site (T600 and S622) of Sld3. Furthermore, our SCMG–dsDNA model revealed that Sld3CBD on CMG appears to contact an N-terminal helix of Mcm2CTD and Sld3CTD could extend to bind to Mcm4NTD through interaction with the NTDs of Mcm2, and 6, rather than the CTD tier (Supplementary Figure 16), indicating that the binding of Sld3–Sld7 to MCM does not appear to affect the ability of AAA+ motors to open and close the MCM ring with a gap between Mcm2CTD and Mcm5CTD [27 [↗](#), 28 [↗](#), 35 [↗](#), 38 [↗](#)]. Taking our findings and those of previous studies together, we propose a detailed process for helicase CMG formation from inactive MCM, as depicted in **Figure 5 A–C** [↗](#), in which Sld7–Sld3 brings Cdc45 into the MCM as a Sld7–Sld3–Cdc45 dimer (Cdc45–Sld3–[Sld7]₂–Sld3–Cdc45), which remains until GINS loading.

The following inquiry concerns the dissociation of Sld3 and other factors. Interestingly, the mutant analysis showed that disrupting just one binding site between Sld3CBD and Cdc45 could cause the dissociation of Sld3CBD and Cdc45, indicating that the binding between Sld3CBD and Cdc45 could be broken easily. Our binding analysis of ssARS1 fragments to Sld3CBD, Sld3CBD–Cdc45, and Sld7–Sld3ΔC–Cdc45 demonstrated the sequence specificity to ssARS1-2 and ssARS1-5 fragments, not others including dsARS1. Considering that ssARS1-2 and ssARS1-5 are on both sides of MCM DH-bound dsDNA at replication origin [39 [↗](#)], the origin unwinding by CMG generates ssDNA and further sequesters the Sld7–Sld3 complex onto ssDNA to remove Sld7–Sld3 from CMG. As a bridge protein, Sld3 recruits Cdc45 to MCM, and phosphorylated Sld3 regulates the recruitment of GINS [40 [↗](#)]. Finally, the release of Sld3 and Sld7 from CMG could be associated with the binding of ssARS1 unwound by CMG and may also be related to the dissociation of Dpb11–Sld2 from CMG (**Figure 5 D, E** [↗](#)) [29 [↗](#), 41 [↗](#)]. Further structural investigation of Sld3–Sld7–Cdc45–MCM in the GINS-recruiting stage is required to validate the complete CMG formation process.

In conclusion, our structural and biochemical studies of Sld3CBD–Cdc45 revealed a detailed process of CMG formation and the subsequent release of the central regulator Sld3 and other factors associated with single-stranded DNA, leading to a deeper understanding of the initialization mechanism of DNA replication.

Materials and Methods

Preparation of proteins

The C-terminal His-tagged Sld3CBD of *S. cerevisiae* was expressed in *Escherichia coli* and prepared as previously described [23 [↗](#)]. For co-expression of *S. cerevisiae* Sld3CBD (S148–K430) and Cdc45 (M1–L650) (Sld3CBD–Cdc45) in *E. coli*, the Sld3CBD DNA fragment containing the His₆-tag (LEHHHHHH) at the C-terminus and the Cdc45 DNA fragment were cloned in the co-overexpression vector pETDuet-1 between the NcoI and SalI restriction sites and the NdeI and XhoI restriction sites, respectively. The primer sequences for Sld3CBD–Cdc45 are listed in Supplementary Table 1.

To confirm the interactions obtained from the Sld3CBD–Cdc45 structure, we constructed four types of single or multi-mutants for Sld3CBD (Sld3-3S: I352S/I355S/L356S, Sld3-3E: 352E/I355E/L356E, Sld3-2R: D344R/D348R, and Sld3-Y: I352Y), and three types of single or multi-mutants for Cdc45 (Cdc45-IIIS: L522S/L527S/V529S, Cdc45-IIIE: L522E/L527E/V529E, Cdc45-IIS: L637S/L641S, Cdc45-IIE: L637E/L641E, and Cdc45-A: R523A) using the Quick Change site-directed mutagenesis method and the inverse PCR method with pETDuet-1–Sld3CBD–Cdc45 as template DNA. The primer sequences for the mutant strains are listed in Supplementary Table 1.

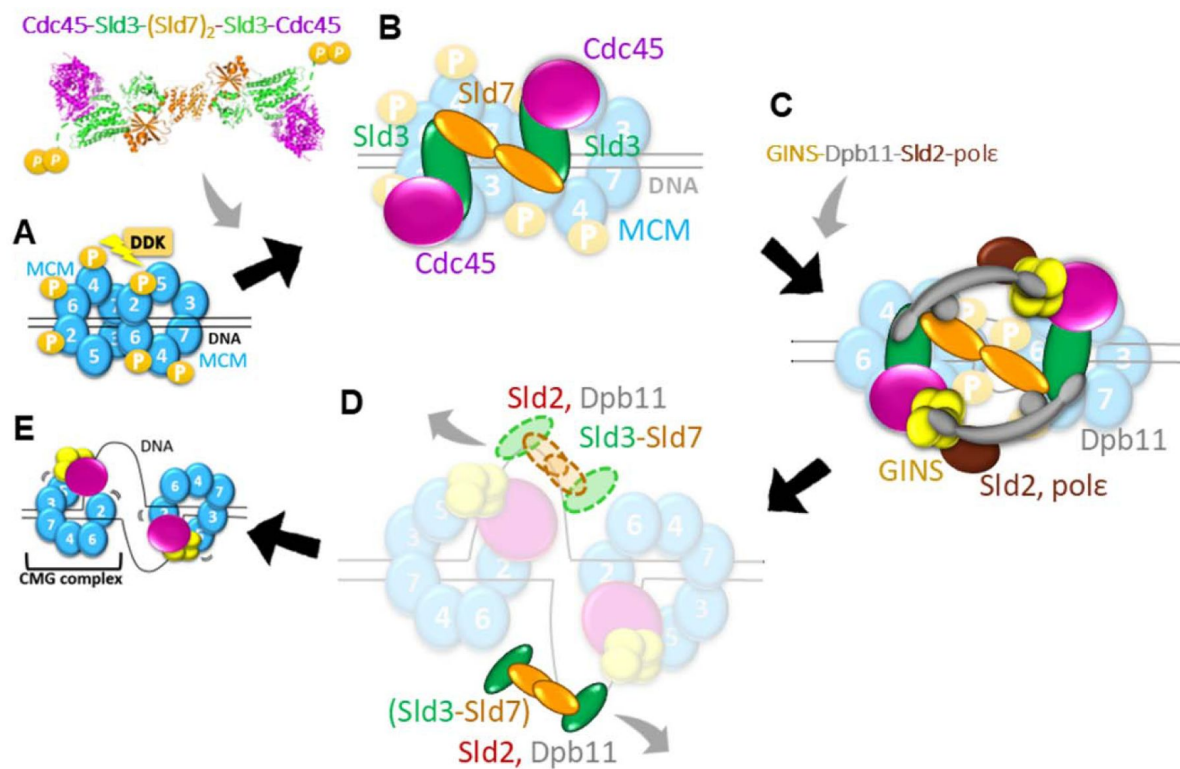


Figure 5.

Proposal for CMG formation with Sld7-Sld3

(A) Phosphorylation of Mcm2,4,6 by DDK after MCM DH was loaded on dsDNA at the replication origin. (B) Cdc45 recruitment to MCM DH by Cdc45-Sld3-[Sld7]₂-Sld3-Cdc45. (C) After CDK-mediated phosphorylation of Sld3CTD in Cdc45-MCM, Dpb11-Sld2 recruits GINS and pole to Sld7-Sld3-Cdc45-MCM to form an active helicase CMG. (D) Unwinding of dsDNA by CMG with MCM DH separation and MCM ring opening. Sld3 and other factors are released upon binding to ssDNA. (E) Each CMG unwinds the dsDNA in two directions, initiating DNA replication.

As *S. cerevisiae* Sld7–Sld3–Cdc45 could not be co-overexpressed in *E. coli*, we attempted to clone it from several other fungal sources. The complex of Sld7 (M1–T268), Sld3ΔC (M1–K430, truncated C-terminal domain), and Cdc45 (M1–I666) from the budding yeast *Kluyveromyces marxianus*, which belongs to the same family as *S. cerevisiae* (Sld7–Sld3ΔC–Cdc45), was cloned into pETDuet-1. We also added a His₆-tag (LEHHHHHH) to the C-terminus of Sld3ΔC. The primer sequences for Sld7–Sld3ΔC and Sld7–Sld3ΔC–Cdc45 are listed in Supplementary Table 1. As Cdc45 mutants can lose the ability to bind to Sld3, we overexpress Sld7–Sld3ΔC by using multi-mutants of Cdc4 (Cdc45-IIS) in Sld7–Sld3ΔC–Cdc45. The Ser-substitution residues (L654S/L658S) of Cdc45-IIS from *K. marxianus* were selected based on sequence conservation. We constructed mutant Sld7–Sld3ΔC–Cdc45-IIS using the Quick Change site-directed mutagenesis method and the inverse PCR method with pETDuet-1–Sld7–Sld3ΔC–Cdc45 as template DNA. The primer sequences for the mutant strains are listed in Supplementary Table 1.

To overexpress Sld3CBD, Sld3CBD–Cdc45, Sld3CBD–Cdc45 mutants, Sld7–Sld3ΔC–Cdc45 and Sld7–Sld3ΔC, the vector of pET26Sld3CBD, pETDuet-1–Sld3CBD–Cdc45, pETDuet-1–Sld3CBD–Cdc45 mutants, pETDuet-1–Sld7–Sld3ΔC–Cdc45 or pETDuet-1–Sld7–Sld3ΔC was transformed into *E. coli* strain BL21 (DE3) through electroporation, followed by preculturing in 5 ml of Luria–Bertani medium containing 100 μg/mL ampicillin at 310 K overnight. The culture was then transferred to 3 L Luria–Bertani medium and grown until the OD₆₀₀ reached 0.6. After cooling the culture for 30 min on ice, overexpression of each sample was induced by the addition of isopropyl-β-D-1-thiogalactopyranoside to a final concentration of 0.5 mM, and then cells were grown for an additional 12 h at 293 K. Cells were harvested by centrifugation at 4,000 *g* for 20 min at 283 K and then resuspended in a buffer containing 20 mM Tris-HCl pH 7.5, 300 mM NaCl, 10% glycerol, 0.1 mg/ml DNase, and 1× protease-inhibitor cocktail (cOmplete EDTA-free; Roche, Basel, Switzerland).

The harvested cells were crushed through sonication and centrifuged at 40,000 *g* for 30 min at 283 K. After filtration through a 0.45-μm filter (Sigma-Aldrich/Merck Millipore, Burlington, MA, USA), the supernatant was then loaded onto the HisTrap HP column equilibrated with buffer A [20 mM Tris-HCl, pH 7.5, 300 mM NaCl, 10% glycerol], and the column was washed with buffer A. The His-tagged target protein was eluted with imidazole at 20% (100 mM), 30% (150 mM), and 50% (250 mM), followed by a linear gradient of 250–500 mM imidazole in buffer B [20 mM Tris-HCl, pH 7.5, 300 mM NaCl, 10% glycerol, 500 mM imidazole]. We analyzed the fractions using sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE). Except for mutants of Sld3CBD–Cdc45, the pooled fractions were then purified through size-exclusion chromatography using a buffer A-equilibrated Superdex 200 16/60 column (GE HealthCare, Chicago, IL, USA). Protein purity was confirmed through SDS-PAGE. The purified samples were concentrated to 10 mg/ml and stored at 193 K.

Crystallization and diffraction data collection

The crystallization screening of Sld3CBD–Cdc45 was performed at 293 K using the sitting-drop vapor diffusion method. The drop was a mixture of 1.0 μl of a 10 mg/ml protein solution with the equivalent volume of reservoir buffer from the commercial crystallization kits (JCSG core I-IV, classics, classics II, PEGs, PEGsII, and MPD suite from Qiagen, Venlo, the Netherlands). The initial needle- and plate-shaped crystals appeared under eight and one conditions, respectively. After optimizing the conditions by altering the type and concentration of precipitant, the salt reagent, and the pH of the buffer, the best crystals were obtained in a solution of 0.2M sodium acetate, 0.1M Bis-Tris propane (pH 6.5: pH 8.5 = 3:7), 20%(w/v) PEG3500, and a drop containing 2.0 μl of a 10 mg/ml protein solution mixed with the equivalent volume of reservoir buffer.

Diffraction data were collected using a beamline BL-17A at the Photon Factory, Tsukuba, Japan. Before data collection, the crystals were cryoprotected by soaking them in a reservoir buffer supplemented with 5% (v/v) glycerol and then flash-cooled under a nitrogen gas stream at 100 K. XDS/XSCALE was used to index, integrate, scale, and merge the dataset [42]. Supplementary Table 2 summarizes the data collection and data processing statistics.

Structure determination and refinement

The structure of Sld3CBD–Cdc45 was determined using the molecular replacement method with the *AutoMR* wizard in *Phenix* [43], [44] and human Cdc45 (PDBID: 5DGO) [31] and Sld3CBD (PDBID:3WI3) [23] structures were used as search models. Several rounds of refinement were performed using the *Phenix_refine* program in *Phenix*, interleaved with manual building and fitting according to the electron density maps of 2Fo-Fc and Fo-Fc using the *Coot* program [44], [45]. Supplementary Table 2 shows the final refinement statistics and geometry defined by MolProbity [46]. All structural diagrams were generated using *PyMol* [47].

Mutant analysis of Sld3 and Cdc45

To analyze the binding sites of Sld3CBD–Cdc45, in conjunction with Cdc45 binding sites to MCM and GINS, we performed co-express pull-down binding assay. We constructed four variants of Sld3CBD and five variants of Cdc45 according to the binding information from the obtained Sld3CBD–Cdc45 structure. We co-express all mutations of Sld3CBD–Cdc45 (Sld3CBD-C-histag) and load them onto the HisTrap HP column under the same conditions as in [Preparation of proteins]. After extracting the samples through Ni-affinity chromatography, we concentrated each eluted sample to 0.005 mg/mL (by nanodrop A280) by centrifugation and checked the binding status of the mutants using SDS-PAGE. Both Sld3CBD and Cdc45 should be observed in the elution group if they formed a complex. The overexpress level of the Cdc45 was checked by - IPTG and +IPTG.

To check whether mutations affected the structure, we performed circular dichroism (CD) spectrometry measurement of Sld3mutants–Cdc45. CD spectra were collected using a J-805 spectropolarimeter (JASCO, Tokyo, Japan) in a quartz cell with a path length of 1 mm in an atmosphere of N₂ at 298K. For CD measurements, the samples were dialyzed in a buffer [20 mM Tris-HCl, pH 7.5, 50 mM NaCl] and adjusted to a concentration of 0.5 mg/mL through absorption. CD spectra for the wavelength range of 190–300 nm were obtained by averaging the results of four scans. The results are given in molar ellipticity per residues $[\theta]_{\text{mrw}} (\times 10^{-3})$ vs the wavelength/nm [48]. The secondary structures of each sample were estimated using the K2D3 method [49]. For wild-type Sld3CBD, we calculated secondary structures from the obtained Sld3CBD–Cdc45 structure.

Growth of mutant cells

The isolation and plasmid shuffling of the temperature-sensitive yeast strain YYK13 (for Sld3 mutant analysis) and Cdc45-35 (for Cdc45 mutant analysis) have been described in a previous study [12]. To investigate Sld3 mutants, strain YYK13 was transformed with the YCplac22 plasmid containing multiple variants of *SLD3* mutant genes. The transformants were streaked on SD plates lacking Trp and Leu (SD-Trp, Leu) and SD-Trp, Leu containing 5-fluoroorotic acid (5-FOA-Trp, Leu), and then incubated at 298 K for 3 days. To analyze the Cdc45 mutant, strain Cdc45-35 containing the *CDC45* mutant gene was re-transformed with the YEplac195 plasmid containing *SLD3* and the YEplac122 plasmid containing *SLD7*. The transformants were streaked on yeast extract–peptone–dextrose plates and incubated at 298 or 305 K for 3 days.

Modelling of complexes

We constructed the models of Sld3CBD–Cdc45–MCM–dsDNA (Sld3CBD–Cdc45–MCM dimer complexed with dsDNA) and SCMG–dsDNA (SCMG dimer complexed with dsDNA) using a two-step process. First, the structure of the MCM–NTDs in the CMG monomer (PDBID: 3JC6) [28] was superimposed on that of MCM DH complexed with dsDNA (5BK4) [35] to obtain a dimer of Cdc45–MCM–GINS complexed with dsDNA (Cdc45–MCM–GINS–dsDNA). Next, the models of Sld3CBD–Cdc45–MCM–dsDNA and SCMG–dsDNA were obtained by superimposing Cdc45NTD from Sld3CBD–Cdc45 onto Cdc45–MCM–GINS–dsDNA.

Dynamic light scattering

We estimated the particle size of the Sld7–Sld3ΔC–Cdc45 complex using dynamic light scattering in terms of peak size [50]. To avoid the effect of concentration, we measured four samples of the Sld7–Sld3ΔC–Cdc45 complex. Each sample was overexpressed independently and purified through size-exclusion chromatography. All samples were concentrated to 10 mg/ml, and stored at 193 K. Before measurement, protein samples were dialyzed in a buffer containing 20 mM Tris-HCl pH 7.5 and 300 mM NaCl and then filtered through a 0.22-μm filter. The pure buffer was measured as a background control and 10 μM lysozyme was measured as a standard control. Dynamic light scattering data were collected and analyzed using a Malvern Zetasizer Nano-ZS instrument (Malvern Panalytical, Malvern, UK) for 0.2–0.5 mg/ml protein samples 500 μl in a microquartz cuvette (Malvern-ZEN0112). An automatic duration model was used to collect data. The Zetasizer 6.20 was utilized for data analysis from three measurements to estimate the particle size of each sample.

Electrophoretic mobility shift assay for ssDNA binding

Sld3 binds to single strands of DNA (ssARS1-2 and ssARS1-5) [26]. To determine the specificity of the ssDNA binding affinity of Sld3, we conducted an electrophoretic mobility shift assay (EMSA) using non-denaturing PAGE (native-PAGE) [51]. Given that Cdc45 binds ssDNA with a nonspecific sequence at lengths greater than 60 bases [36], we designed three fragments of ssDNA 40 bases in length (first half 1–40 bp, second half 41–80 bp, and middle half 21–60 bp) for each ssDNA 80-bp segment. All ssDNA fragments of *S. cerevisiae* were synthesized by Sigma-Aldrich. The 40-bp dsDNA fragments (dsARS1-34_1:ssARS1-3_1/ssARS1-4_2) were converted by annealing them using the PCR protocol and then checked through polyacrylamide gel electrophoresis. The loaded samples were incubated overnight at 277 K in TMK buffer (20mM Tris-HCl pH8, 100mM MgCl₂, 200mM KCl) containing synthesized ssDNA and varying amounts of proteins at ssDNA: protein molecular ratios of 1:0, 1:1, 1:2, 1:4, and 0:1 with 10 pM as 1 unit. After incubation, the mixtures were loaded onto a polyacrylamide gel (5% (w/v) acrylamide (39:1), 10% 10 × running buffer (0.25 M Tris, 1.92 M Glycine), 0.1% Ammonium peroxodisulphate, 0.06% (v/v) TEMED) without denaturing (native-PAGE). EMSA was performed at 10 mA/200 V per gel for 40 min at 277 K in 1 × running buffer. After electrophoresis, the reaction products were visualized using SYBR safe (SYBR safe:1 × running buffer = 0.0001:1) to stain the ssDNA and Coomassie brilliant blue (G-250) to stain the proteins. We repeated the EMSA experiments three or more times to confirm the readability. Considering the functional similarity of ARS1-core, the EMSA of Sld7–Sld3ΔC and Sld7–Sld3ΔC–Cdc45 of *K. marxianus* used ssDNA fragments of *S. cerevisiae*.

Data Availability

Structures and associated experimental data for Sld3CBD-Cdc45 complex (8J09) were deposited in the Protein Data Bank Japan.

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Additional information

Author Contributions

Hao Li: Investigation, Formal analysis, Writing—original draft & editing. Izumi Ishizaki: Investigation, Formal analysis. Koji Kato: Formal structure analysis. XiaoMei Sun: Investigation. Sachiko Muramatsu: Investigation. Hiroshi Itou: Investigation, Writing—review & editing. Toyoyuki Ose: Formal analysis. Hiroyuki Araki: conceptualization, Formal analysis, Writing—review & editing. Min Yao: conceptualization, Formal analysis, Methodology, Writing—original draft & editing.

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Additional files

Supplemental Figures and Tables [↗](#)

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Reviewer #1 (Public review):

Summary:

The crystal structure of the Sld3CBD-Cdc45 complex presented by Li et al. is a significant contribution that enhances our understanding of CMG formation during the rate-limiting step of DNA replication initiation. This structure provides crucial insights into the intermediate steps of CMG formation, and the particle analysis and model predictions compellingly describe the mechanism of Cdc45 loading. Building upon previously known Sld3 and Cdc45 structures, this study offers new perspectives on how Cdc45 is recruited to MCM DH through the Sld3-Sld7 complex. The most notable finding is the structural rearrangement of Sld3CBD upon Cdc45 binding, particularly the α 8-helix conformation, which is essential for Cdc45 interaction and may also be relevant to its metazoan counterpart, Treslin. Additionally, the conformational shift in the DHHA1 domain of Cdc45 suggests a potential mechanism for its binding to Mcm2NTD. Furthermore, Sld3's ssDNA-binding experiments provide evidence of its novel functions in the DNA replication process in yeast, expanding our understanding of its role beyond Cdc45 recruitment.

Strengths:

The manuscript is generally well-written, with a precise structural analysis and a solid methodological section that will significantly advance future studies in the field. The predictions based on structural alignments are intriguing and provide a new direction for exploring CMG formation, potentially shaping the future of DNA replication research. This research also opens up several new opportunities to utilize structural biology to unravel the molecular details of the model presented in the paper.

Weaknesses:

The main weakness of the manuscript lies in the lack of detailed structural validation for the proposed Sld3-Sld7-Cdc45 model, and its CMG bound models, which could be done in the future using advanced structural biology techniques such as single particle cryo-electron microscopy. It would also be interesting to explore how Sld7 interacts with the MCM helicase, and this would help to build a detailed long-flexible model of Sld3-Sld7-Cdc45 binding to MCM DH and to show where Sld7 will lie on the structure. This will help us to understand how Sld7 functions in the complex. Also, future experiments would be needed to understand the molecular details of how Sld3 and Sld7 release from CMG is associated with ssARS1 binding.

<https://doi.org/10.7554/eLife.101717.2.sa3>

Reviewer #2 (Public review):

Summary

The manuscript presents valuable findings, particularly in the crystal structure of the Sld3CBD-Cdc45 interaction and the identification of additional sequences involved in their binding. The modeling of the Sld7-Sld3CBD-CDC45 subcomplex is novel, and the results provide insights into potential conformational changes that occur upon interaction. Although the single-stranded DNA binding data from Sld3 of different species is a minor weakness, the experiments support a model in which the release of Sld3 from the complex may be promoted by its binding to origin single-stranded DNA exposed by the helicase.

Strengths

- The Sld3CBD-Cdc45 structure is a novel contribution, revealing critical residues involved in the interaction.
- The model structures generated from the crystal data are well presented and provide valuable insights into the interaction sequences between Sld3 and Cdc45.
- The experiments testing the requirements for interaction sequences are thorough and conducted well, with clear figures supporting the conclusions.
- The conformational changes observed in Sld3 and Cdc45 upon binding are interesting and enhance our understanding of the interaction.
- The modeling of the Sld7-Sld3CBD-CDC45 subcomplex is a new and valuable addition to the field.
- The proposed model of Sld3 release from the complex through binding to single stranded DNA at the origin is intriguing.

Weaknesses

- The section on the binding of Sld3 complexes to origin single-stranded DNA is somewhat weakened by the use of Sld3 proteins from different species. The comparisons between Sld3-CBD, Sld3CBD-Cdc45, and Sld7-Sld3CBD-Cdc45 involve complexes from different species, limiting the comparisons' value.
- Although the study reveals that Sld3 binds to different residues of Cdc45 than those previously shown to bind Mcm or GINS, the data in the paper do not shed any additional light

on how GINS and Sld3 binding to Cdc45 or Mcms. would affect each other. Other previous research has suggested that the binding of GINS and Sld3 to Mcm or Cdc45 may be mutually exclusive. The authors acknowledge that a structural investigation of Sld3, Sld7, Cdc45, and MCM during the stage of GINS recruitment will be a significant goal for future research.

<https://doi.org/10.7554/eLife.101717.2.sa2>

Reviewer #3 (Public review):

Summary:

The paper by Li et al. describes the crystal structure of a complex of Sld3-Cdc45-binding domain (CBD) with Cdc45 and a model of the dimer of an Sld3-binding protein, Sld7, with two Sld3-CBD-Cdc45 for the tethering. In addition, the authors showed the genetic analysis of the amino acid substitution of residues of Sld3 in the interface with Cdc45 and biochemical analysis of the protein interaction between Sld3 and Cdc45 as well as DNA binding activity of Sld3 to the single-strand DNAs of the ARS sequence.

Strengths:

The authors provided a nice model of an intermediate step in the assembly of an active Cdc45-MCM-GINS (CMG) double hexamers at the replication origin, which is mediated by the Sld3-Sld7 complex. The dimer of the Sld3-Sld7 complexes tethers two MCM hexamers together for the recruitment of GINS-Pol epsilon on the replication origin.

Weaknesses:

The biochemical analysis should be carefully evaluated with more quantitative ways to strengthen the authors' conclusion even in the revised version.

<https://doi.org/10.7554/eLife.101717.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

The crystal structure of the Sld3CBD-Cdc45 complex presented by Li et al. is a novel contribution that significantly advances our understanding of CMG formation during the rate-limiting step of DNA replication initiation. This structure provides insights into the intermediate steps of CMG formation. The study builds upon previously known structures of Sld3 and Cdc45 and offers new perspectives into how Cdc45 is loaded onto MCM DH through Sld3-Sld7. The most notable finding is the structural difference in Sld3CBD when bound to Cdc45, particularly the arrangement of the α 8-helix, which is essential for Cdc45 binding and may also pertain to its metazoan counterpart, Treslin. Additionally, the conformational shift in the DHHA1 domain of Cdc45 suggests a possible mechanism for its binding to MCM2NTD.

Strengths:

The manuscript is generally well-written, with a precise structural analysis and a solid methodological section that will significantly advance future studies in the field. The predictions based on structural alignments are intriguing and provide a new direction for exploring CMG formation, potentially shaping the future of DNA replication research.

Weaknesses:

The main weakness of the manuscript lies in the lack of experimental validation for the proposed Sld3-Sld7-Cdc45 model. Specifically, the claim that Sld3 binding to Cdc45-MCM does not inhibit GINS binding, a finding that contradicts previous research, is not sufficiently substantiated with experimental evidence. To strengthen their model, the authors must provide additional experimental data to support this mechanism. Also, the authors have not compared the recently published Cryo-EM structures of the metazoan CMG helicases with their predicted models to see if Sld3/Treslin does not cause any clash with the GINS when bound to the CMG. Still, the work holds great potential in its current form but requires further experiments to confirm the authors' conclusions.

We appreciate the reviewers' careful reading and the comments.

Our structural analysis of Sld3CBD-Cdc45 showed the detailed interaction map between Sld3CBD and Cdc45 at 2.6 Å resolution. The Sld3, MCM and GINS binding sites of Cdc45 completely differed, suggesting that the Sld3CBD, Cdc45 and GINS could bind to MCM together. The SCMG-DNA model confirmed such a binding manner, although our study does not show how this binding manner affects the GINS loading by other initiation factors (Dpb11, Sld2, et. al). Regarding the previous studies, competition of Sld3 and GINS for binding to Cdc45 or Cdc45-MCM (Bruck et. al), which may be caused by the conformation change of Cdc45 DHHA1 between Sld3CBD-Cdc45 and CMG. We modified our manuscript and discussed (P7/L168-173, and P10/L282-286). Following the comment, we checked the recently published Cryo-EM structure (PDBID:8Q6O) with their predicted models of the metazoan CMG helicases (P7/L198-P8/L202) and added the Cdc45 mutation experiments to confirm our conclusion ([Recommendations for the authors] Q18).

Reviewer #2 (Public review):

Summary

The manuscript presents valuable findings, particularly in the crystal structure of the Sld3CBD-Cdc45 interaction and the identification of additional sequences involved in their binding. The modeling of the Sld7-Sld3CBD-CDC45 subcomplex is novel, and the results provide insights into potential conformational changes that occur upon interaction. However, the work remains incomplete as several main claims are only partially supported by experimental data, particularly the proposed model for Sld3 interaction with GINS on the CMG. Additionally, the single-stranded DNA binding data from different species do not convincingly advance the manuscript's central arguments.

Strengths

(1) The Sld3CBD-Cdc45 structure is a novel contribution, revealing critical residues involved in the interaction.

(2) The model structures generated from the crystal data are well presented and provide valuable insights into the interaction sequences between Sld3 and Cdc45.

(3) The experiments testing the requirements for interaction sequences are thorough and conducted well, with clear figures supporting the conclusions.

(4) The conformational changes observed in Sld3 and Cdc45 upon binding are interesting and enhance our understanding of the interaction.

(5) The modeling of the Sld7-Sld3CBD-CDC45 subcomplex is a new and valuable addition to the field.

Weaknesses

(1) The proposed model for Sld3 interacting with GINS on the CMG needs more experimental validation and conflicts with published findings. These discrepancies need more detailed discussion and exploration.

Our structural analysis experiment of Sld3CBD-Cdc45 showed the detailed interaction information between Sld3CBD and Cdc45 at 2.6 Å resolution. The Sld3CBD-binding site of Cdc45 is completely different from that of GINS and MCM binding to Cdc45, suggesting that the Sld3CBD, Cdc45, and GINS could bind to MCM together. The SCMG-DNA model confirmed such a binding manner. Following the comment, we added a Cdc45 mutant analysis, disrupting the binding to MCM and GINS but not affecting the Sld3CBD binding (Supplementary Figure 9). Our model is consistent with the GINS-loading requirement (the phosphorylation of Sld3 on Cdc45-MCM) and has no discrepancies with the stepwise loading fashion (Please see the responses to [Recommendations for the authors] Reviewer#1-Q14-15)). Regarding the previous studies, competition of Sld3 and GINS for binding to Cdc45 or Cdc45-MCM (Bruck et. al), by *in vitro* binding experiments, please see the responses to [Recommendations for the authors] Q6.

(2) The section on the binding of Sld3 complexes to origin single-stranded DNA needs significant improvement. The comparisons between Sld3-CBD, Sld3CBD-Cdc45, and Sld7-Sld3CBD-Cdc45 involve complexes from different species, limiting the comparisons' value.

As suggested, we tried to improve the ssDNA-binding section (Please see the responses to [Recommendations for the authors]: Q4 and Q5). We used Sld7-Sld3CBD-Cdc45 from different sources due to limitations in protein expression. These two sources belong to the same family and the proteins Sld7, Sld3 and Cdc45 have sequence conservation with similar structures predicted by the alphafold3 (RMSD = 0.356, 1.392, and 0.891 for Ca atoms of Sld7CTD, Sld7NTD-Sld3NTD, and Sld3CBD-Cdc45). Such similarity in source and protein lever allows us to do the comparison.

(3) The authors' model proposing the release of Sld3 from CMG based on its binding to single-stranded DNA is unclear and needs more elaboration.

Considering that ssDNA (ssARS1) is produced by CMG, the ssDNA-binding of Sld3 should happen after forming an active CMG. Therefore, the results of ssDNA binding experiments implied that the Sld3 release could be with the binding to ssDNA produced by CMG. We tried to present more elaborations in the revised version. (Please see the responses to [Recommendations for the authors] Q4, Q5).

Reviewer #3 (Public review):

Summary:

The paper by Li et al. describes the crystal structure of a complex of Sld3-Cdc45-binding domain (CBD) with Cdc45 and a model of the dimer of an Sld3-binding protein, Sld7, with two Sld3-CBD-Cdc45 for the tethering. In addition, the authors showed the genetic analysis of the amino acid substitution of residues of Sld3 in the interface with Cdc45 and

biochemical analysis of the protein interaction between Sld3 and Cdc45 as well as DNA binding activity of Sld3 to the single-strand DNAs of the ARS sequence.

Strengths:

The authors provided a nice model of an intermediate step in the assembly of an active Cdc45-MCM-GINS (CMG) double hexamers at the replication origin, which is mediated by the Sld3-Sld7 complex. The dimer of the Sld3-Sld7 complexes tethers two MCM hexamers together for the recruitment of GINS-Pol epsilon on the replication origin.

Weaknesses:

The biochemical analysis should be carefully evaluated with more quantitative ways to strengthen the authors' conclusion.

We thank your positive assessment. We provided more quantitative information and tried to quantify the experiments as suggested (Please see the responses to [Recommendations for the authors]).

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

I have several concerns that I will outline below, accompanied by my suggestions.

(1) "The title of the paper- "Structural and functional insights into Cdc45 recruitment by Sld7-Sld3 for CMG complex Formation," appears misleading because it appears that authors present a structure of Sld3-Sld7 in complex with Cdc45, which is not the case here. If authors can provide additional structures proving the function of this complex, then this title justifies it. Otherwise, I recommend making a title that justifies the presented work in its current form.

Following the comment, we change the title to “Sld3CBD-Cdc45 structural insights into Cdc45 recruitment for CMG complex formation”.

(2) In lines 70-72, where the authors mention the known structures of different proteins, intermediates, and complexes, I recommend including PDB IDs of the described structures and reference citations. This will help the readers to analyze what is missing in the pathway and why this structure is essential.

Following the comment, we added PDBIDs and references (P3/L72-74).

(3) The representation of Figure 1A is unclear and looks clumsy. If the structure were rotated in another orientation, where $\alpha 8$ and $\alpha 9$ would be displayed on the forward side, it would be more helpful to understand the complex forming regions by looking at the structure. Also, I recommend highlighting the $\alpha 8$ and $\alpha 9$ in a contrasting color to be easily visible and attract readers' attention. Similarly, it would also be helpful if DHAA1 would be shown in a different color.

Following the comment, we modified the Figure1 to show $\alpha 8$ and $\alpha 9$ of Sld3CBD and DHAA1 of Cdc45 clearly in revised version.

(4) Can authors add a supplementary figure showing the probability of disorderiness of the $\alpha 8$ helix region in the Sld3? Also, highlight what region became ordered in their structure.

Yes, we have showed the disordered $\alpha 8$ helix region and highlight ordered $\alpha 8$ in the Sld3 in Figure S4 A.

(5) Can you compare the Cdc45 long distorted helix (Supplementary Figure 3B) in the Sld3-Cdc45 complex with the *Xenopus* and *Drosophila* Cdc45 from their CMG structures? Also, can the authors explain why this helix is destabilized in their structure but is relatively stable in another Cdc45 structure (in CMG and HuCdc45)?

We have checked all Cdc45 from published cryo-EM CMG structures, including *Xenopus* CMG-donson (8Q6O) and *Drosophila* CMG (6RAW), and all of them ordered the long helix in the CMG complex, whereas this long helix was disordered in the crystal structure of Sld3CBD-Cdc45 and *Entamoeba histolytica* Cdc45. The crystal packing around the long helix showed that it looks to be stabilized by crystal packing only in huCdc45, therefore we suggested that this long helix is detestable for crystallization.

(6) I recommend adding the following parameters to Supplementary Table 2: 1. Rmerge values, 2. Wilson B factor, 3. Average B factor, and 4. Total number of molecules in ASU.

We are sorry to make a mistake about Rmerge in Table 2. We correct it. We added the Wilson B factor, the average B factor, and the total number of Sld3CBD-Cdc45 in ASU.

(7) Can authors provide the B factor values of the $\alpha 8$ helix of Sld3?

We checked the B factor values of the helix $\alpha 8$ CTP of Sld3 in Sld3CBD-Cdc45. Since this helix binds to Cdc45 stably, the average B factor of the main chain is $45 \text{ \AA}^2$ less than that of the whole structure. We added the average B factor of helix $\alpha 8$ CTP into the Supplementary Figure 4A legend.

(8) Can authors explain why higher Ramachandran outliers exist in their structure? Can it be reduced below 1% during refinement?

There are 13 outliers (1.67%) in different places: four are close to the disorder regions (poor electron map), four are in a loop with poor map and the remains are turn parts or a loop. For the residues with poor electron maps, we could not modify them to the allow Ramachandran region with low Rfree value, so we could not reduce them to below 1% during refinement while keeping the current Rfree value.

(9) In Supplementary Figure 8, please show the CD spectra of the Sld3WT. Why is the Sld3-3S peak relatively flat? Was the sample precipitating while doing the measurements, or does it have less concentration than others?

To check the folding of the mutants, we did CD experiments with the estimated secondary structure elements. Because WT Sld3CBD was prepared in a complex with Cdc45, while the mutants of Sld3CBD existed along, we calculated the elements of secondary structure from the crystal structure of Sld3CBD-Cdc45. The concentration of samples was controlled to the same level for CD measurement. The relative plat of the Sld3-3S peak may be caused by precipitating while doing the measurement.

(10) Can authors generate the alpha fold three models of the Sld3CBD-Cdc45-MCM-dsDNA and SCMG-dsDNA and compare them with the models they have generated?

We tried to predict the Sld3CBD-Cdc45-MCM-dsDNA and SCMG-dsDNA using Alphafold3. Although the results showed similar structures to our models, many parts were disordered.

So, we did not use the predicted structures.

(11) The authors say that the overall molecular mass of the Sld7-Sld3ΔC-Cdc45 was >400kDa on the SEC column. However, the column used for purifying this complex and the standards that were run on it for molecular weight calculations have not been written anywhere. If the Superdex 200 column was used, then the sample of more than 400kDa should not elute at the position shown in Supplementary Figure 2B. I recommend showing the standard MW plot and where the elution volume of the Sld7-Sld3ΔC-Cdc45 lies on the standard curve. Also, add how molecular weight calculations were done and the calculated molecular mass.

Following the comment, we added a measurement of Superdex 200 16/60 column (SEC) using a standard sample kit into Supplementary Figure 2 to show that the molecular weight of the peak at the position was estimated to be > 400 k Da.

(12) I also recommend using at least one of the techniques, either SEC-MALS or AUC, to calculate the actual molecular mass of the Sld7-Sld3ΔC-Cdc45 complex and to find its oligomeric state. If the authors want to prove their hypothesis that a dimer of this complex binds to MCMDH, it is essential to show that it exists as a dimer. Based on the current SEC profile, it appears as a monomer peak if the S200 SEC column is being used.

As the response to (11), we added the standard MW plot (measurement using Superdex 200 16/60 column) using a standard sample kit. The molecular weight at the peak elution position of Sld7-Sld3ΔC-Cdc45 was estimated to be 429k Da. Considering that the Sld7-Sld3ΔC-Cdc45 dimer should be a flexible long-shaped molecule, the elution position could be at a larger molecular weight position than the real one (158 x 2 k Da). We also tried to confirm the particle size using SEC-SAXS, as the response to the next question (13).

(13) Dynamic light scattering is not the most accurate method for calculating intermolecular distance. I recommend using another technique that calculates the accurate molecular distances between two Cdc45 if Sld7-Sld3ΔC-Cdc45 is forming a dimer. Techniques such as FRET could be used. Otherwise, some complementary methods, such as SAXS, could also be used to generate a low-resolution envelope and fit the speculated dimer model inside, or authors could try negative staining the purified Sld7-Sld3ΔC-Cdc45 and generate 2D class averages and low-resolution ab initio models to see how the structure of this complex appears and whether it satisfies the speculated model of the dimeric complex.

We have tried both negative staining TEM and SEC-SAXS experiments. We could not obtain images good enough of negative staining of TEM to generate 2D class averages and low-resolution ab initio models. The results of SEC-SAXS provided a molecular weight of 370 - 420 kDa, and an $R_g > 85 \text{ \AA}$, which are consistent with our conclusion from SEC and DLS results but with large error due to the measurement temperature at 10-15°C (measuring equipment limitation). The peak of SCE-SAXS under measurement conditions was not as sharp as purification at 4°C and SAXS data is not good enough to make a molecular model, so we did not add them to our manuscript.

(14) Authors mentioned in the introduction section (lines 72-73) that based on the single-molecule experiments, Cdc45 is recruited in a stepwise manner to MCMDH. If this is true and if Sld7-Sld3ΔC-Cdc45 forms a dimer, this is also true, then for stepwise recruitment, the dimer will have to break into monomers, and this will be an energy-expensive process for the cell. So, would such a process occur physiologically? Can the authors explain how this would physiologically happen inside the cell?

Sld7-Sld3-Cdc45 consists of domains linked by long loops, so the dimer Cdc45-Sld3-[Sld7]2-Sld3-Cdc45 is flexible long-sharp. Such a flexible dimer does not mean that two Cdc45 molecules must bind to MCM DH simultaneously and may bind to MCM DH by stepwise manner. The dimer formation of Sld7-Sld3-Cdc45 is advantageous for recruiting efficiently and saving energy. Moreover, our proposal of Cdc45-Sld3-[Sld7]2-Sld3-Cdc45 on MCM DH could be a stage during CMG formation in the cell. Following the comment, we added such descriptions (P7/L194, and P10/L276-279).

(15) Can authors show experimentally that a dimer of Sld7-Sld3ΔC-Cdc45 is binding to MCMDH and not a monomer in a stepwise fashion?

In our study, we provided experiments of particle size to show the dimer of Sld7-Sld3-Cdc45 off MCM DH and a model of SCMG to indicate the dimer of Sld7-Sld3ΔC-Cdc45 on MCM DH. This question should be addressed future by the Cryo-EM of Sld7-Sld3-Cdc45-MCM DH or Sld7-Sld3-CMG. As the response to Q14, the flexible dimer of Sld7-Sld3ΔC-Cdc45 binding on MCMDH does not contradict the stepwise-loading fashion. The dimer of Sld7-Sld3ΔC-Cdc45 binding on MCM DH shows a stage.

(16) Can authors highlight where Sld7 will lie on their model shown in Figures 3A and 3C, considering their model shown in 3B is true?

We predict that the Sld7-Sld3-Cdc45 should be in a dimer form of Cdc45-Sld3-[Sld7]2-Sld3-Cdc45 based on the structures and the particle size analysis. The Sld7 dimer could be across MCM DH on the top of Figure 3A right and 3C right. However, we could not add the Sld7 molecule to the models because there is no interaction data between Sld7 and MCM.

(17) In Supplementary Figure 10, can authors show the residues between the loop region highlighted in the dotted circle to show that there is no steric clash between the residues in that region of their predicted model?

Following the comment, we added the residues in Supplementary Figure 10 (Supplementary Figure 11 in the revised version) to show no steric clash in our predicted model.

(18) It is essential to show experimentally that Sld3CBD neighbors MCM2 and binds Cdc45 on the opposite side of the GINS binding site. I recommend that the authors design an experiment that proves this statement. Mutagenesis experiments for the predicted residues that could be involved in interaction with proper controls might help to prove this point. Since this is the overall crux of the paper, it has to be demonstrated experimentally.

We thank the reviewer's recommendation. Our structural analysis experiment shows the interaction information between Sld3CBD and Cdc45 at 2.6 Å resolution. The Sld3CBD-binding site, GINS-binding site, and MCM-binding site of Cdc45 are completely different, indicating that the Sld3CBD, Cdc45 and GINS could bind to MCM together. The SCMG model confirmed such a binding manner. Following the recommendation, we added mutant analysis of Cdc45 G367D and W481R, which was reported to disrupt the binding to MCM and GINS, respectively. Both mutants do not affect the binding to Sld3CBD as we predicted (Supplementary Figure 9B). We modified our manuscript and discussed this point more clearly (P7/L170-173).

(19) I recommend rewriting the sentence in lines 208-210. During EMSA experiments, new bands do not appear; instead, there is no shift at lower ratios, so you see a band similar to the control for Sld3CBD-Cdc45. So, re-write the sentence correctly to avoid confusion when interpreting the result.

Following the comment, we rewrote this sentence to "The ssDNA band remained (Figure 4B) and new bands corresponding to the ssDNA–protein complex appeared in CBB staining PAGE (Supplementary Figures 13) when the Sld3CBD–Cdc45 complex was mixed with ssDNA at the same ratio, indicating that the binding affinity of Sld3CBD–Cdc45 for ssDNA was lower than that of Sld3CBD alone" (P8/L226-229)

(20) Since CDK-mediated phosphorylation of Sld3 is known to be required for GINS loading, the ssDNA binding affinity of phosphorylated Sld3 remains the same. I wonder what would happen if phosphorylated Sld3 were used for the experiment shown in Figure 4B.

The CDK phosphorylation site is located at Sld3CTD and our ssDNA-binding experiment did not include the Sld3CTD, so phosphorylated Sld3 does not affect the results shown in Figure 4B.

(21) Sld3CBD-Cdc45 has a reduced binding affinity for ss DNA, and Sld7-Sld3ΔC-Cdc45 and S17-Sld3ΔC have a similar binding affinity to Sld3CBD based on figure 4B. It appears that Sld3CBD reduces the DNA binding affinity of CDC45 or vice versa. Is it correct to say so?

Our opinion is "vice versa". Cdc45 reduces the ssDNA-binding affinity of Sld3CBD. Although we could not point out the ssDNA-binding sites of Sld3CBD, the surface charge of Sld3CBD implies that α8CTP could contribute to ssDNA-binding (Supplementary Figures 15).

(22) Cdc45 binds to the ssDNA by itself, but in the case of Sld3CBD-Cdc45, the binding affinity is reduced for Sld3CBD and Cdc45. Based on their structure, can authors explain what leads to this complex's reduced binding affinity to the ssDNA? Including a figure showing how Sld7-Sld3CBD-Cdc45 interacts with the DNA would be a nice idea.

Previous studies showed that Cdc45 binds tighter to long ssDNA (> 60 bases) and the C-terminus of Cdc45 is responsible for the ssDNA binding activity. The structure of Sld3CBD–Cdc45 shows the C-terminal domain DHHA1 of Cdc45 binds to Sld3CBD, which may lead to Sld3CBD–Cdc45 complex reduced ssDNA-binding affinity of Cdc45. We agree that showing a figure of how Sld7–Sld3CBD–Cdc45 interacts with ssDNA is a nice idea. However, there is no detailed interaction information between Sld7–Sld3ΔC–Cdc45 and ssDNA, so we could not give a figure to show the ssDNA-binding manner. We added a figure to show the surface charges of Sld3CBD of Sld3CBD–Cdc45, and Sld3NTD–Sld7NTD, respectively (Supplemental Figure 15).

(23) Based on the predicted model of Sld7-Sld3 and Cdc45 complex, can authors explain how Sld7 would restore the DNA binding ability of the Sld3CBD?

It can be considered that Sld7 and Sld3NTD could bind ssDNA. Although we did not perform the ssDNA-binding assay of Sld7, the Sld3NTD–Sld7NTD surface shows a large positive charge area which may contribute to ssDNA-binding (Supplemental Figure 15). We added the explanation (P9/L245-248).

(24) It would be important to show binding measurements and Kd values of all the different complexes shown in Figure 4B with ssDNA to explain the dissociation of Cdc45 from Sld7-Sld3 after the CMG formation. I also recommend describing the statement from lines 224-227 more clearly how Sld7-Sld3-Cdc45 is loading Cdc45 on CMG.

As the reviewer mentioned, the binding measurements and Kd of values of all the different complexes are important to explain the dissociation of Sld7–Sld3 from CMG. The pull-down

assay using chromatography may be affected by balancing the binding affinity and chromatography conditions. Therefore, we used EMSA with native-PAGE, which is closest to the natural state. However, the disadvantage is that the K_d values could not be estimated. For lines 224-227, the ssARS1-binding affinity of Sld3 and its complex should relate to the dissociation of Sld7-Sld3 from the CMG complex but not Cdc45 loading, because ssARS1 is unwound from dsDNA by the CMG complex after Cdc45 and GINS loading. We modified the description (P9/L248-251).

(25) Can authors explain why SDS-PAGE was used to assess the ssDNA (See line 420)?

We are sorry for making this mistake and corrected it to “polyacrylamide gel electrophoresis”.

(26) In line 421, can the authors elaborate on a TMK buffer?

We are sorry for this omission and added the content of the TMK buffer (P16/L453).

(27) I am curious to know if the authors also attempted to Crystallize the Sld7-Sld3CBD-Cdc45 complex. This complex structure would support the authors' hypothesis in this article.

We tried to crystallize Sld7-Sld3 Δ -Cdc45 but could not get crystals. We also tried using cryo-EM but failed to obtain data.

Reviewer #2 (Recommendations for the authors):

(1) The manuscript would be strengthened if the authors acknowledged in greater detail how their work agrees with or disagrees with Itou et al. (PMID: 25126958 DOI: 10.1016/j.str.2014.07.001). The introduction insufficiently described the findings of that previous work in lines 63-64.

We compared Sld3CBD in Sld3CBD-Cdc45 to the monomer reported by Itou et al. (PMID: 25126958 DOI: 10.1016/j.str.2014.07.001) in the section of [The overall structure of Sld3CBD-Cdc45] and point out the structural similarity and difference (P5/L105-106), especially, conformation change of Sld3CBD α 8 for binding to Cdc45, which agrees to the mutant experiments of Itou et al., (P3/L126-127). Another Cdc45-binding site of Sld3CBD in the Sld3CBD-Cdc45 complex is α 9 not residues predicted in previous studies.

(2) Figure 2. Could you please perform and present data from multiple biological replicates (e.g., at least two independent experiments) for each mutant strain? This would help ensure that the observed pull-downs (2A-B) and growth patterns (2C) are consistent and reproducible.

We have done pull-downs three times from co-expression to purification and pull-down assay. We added descriptions to the method of [Mutant analysis of Sld3 and Cdc45]. The growth patterns are two times in Figure 2C.

(3) Figure 3B. The match between the predicted complex length and particle size measured by dynamic light scattering (DLS) is striking. Did the authors run the analysis with vehicle controls and particle size standards? There is no mention of these controls.

Following the comment, we added the control data of buffer and standard protein lysozyme, and the descriptions to the method of [Dynamic light scattering].

(4) Figure 4. In lines 216-217, the authors write that the binding of the *K. marxianus* complex "demonstrates that the presence of Sld7 could restore the single-stranded DNA binding capacity of Sld3." Another explanation is that complexes from each species bind differently. If the authors want to make a strong claim, they should compare the binding of complexes containing the same proteins.

Agree with the comment, to make a strong claim using samples from the same source is better. Due to limitations in protein overexpression, we used Sld7-Sld3 Δ C-Cdc45 from different sources two sources belong to the identical family (Saccharomycetaceae) and the proteins Sld7, Sld3 and Cdc45 have sequence conservation with similar structures (RMSD = 0.356, 1.392, and 0.891 for Ca atoms of Sld7CTD, Sld7NTD-Sld3NTD, and Sld3CBD-Cdc45) predicted by the alphafold3. Such similarity in source and protein level allows us to do the comparison. Moreover, we modified the description to "indicates that the presence of Sld7 and Sld3NTD could increase the ssDNA-binding affinity to a level comparable to that of Sld3CBD.

(5) The logic of the following is unclear: "Considering that ssDNA is unwound from dsDNA by the helicase CMG complex, Sld7-Sld3 Δ C-Cdc45, and Sld7-Sld3C having a stronger ssDNA-binding capacity than Sld3CBD-Cdc45 may imply a relationship between the dissociation of Sld7-Sld3 from the CMG complex and binding to ssDNA unwound by CMG." (Lines 224-227). How do the authors imagine that the binding affinity difference due to Sld7 contributes to the release of Sld3? Please explain.

Considering that ssARS1 is unwound from dsARS1 by the activated helicase CMG complex formed after loading Cdc45 and GINS, Sld3-Sld7 having a stronger ssARS1-binding affinity may provide an advantage for the dissociation of Sld7-Sld3 from the CMG complex. We modified the sentence of Lines 224-227 (P9/L248-251).

(6) The authors suggest that the release of Sld3 from the helicase is related to its association with single-stranded ARS1 DNA. They refer to the work of Bruck et al. (doi: 10.1074/jbc.M111.226332), which demonstrates that single-stranded origin DNA inhibits the interaction between Sld3 and MCM2-7 *in vitro*. The authors selectively choose data from this previous work, only including data that supports their model while disregarding other data. This approach hinders progress in the field. Specifically, Bruck proposed a model in which the association of Sld3 and GINS with MCM2-7 is mutually exclusive, explaining how Sld3 is released upon CMG assembly. In Figure 3 of the authors' model, they suggest that Sld3 can associate with MCM2-7 through CDC45, even when GINS is bound. Furthermore, Bruck's work showed that ssARS1-2 does not disrupt the Sld3-Cdc45 interaction. Instead, Bruck's data demonstrated that ssARS1-2 disrupts the interaction between MCM2-7 and Sld3 without Cdc45. While we do not expect the authors to consider all data in the literature when formulating a model, we urge them to acknowledge and discuss other critical data that challenges their model. Additionally, it would be beneficial for the field if the authors include both modes of Sld3 interaction with MCM2-7 (i.e., directly with MCM or through CDC45) when proposing a model for how CMG assembly and Sld3 release occurs.

In our discussion, we referred to the studies of Bruck's data (doi: 10.1074/jbc.M111.226332) but did not discuss more because we didn't perform similar experiments *in vitro*, and we do not think that no discussion hinders progress in the field. Promoting research progress, the new experiment should provide a new proposal and updated knowledge. Although we do not know exactly the positional relationship between Sld3 and Dpb11-Sld2 on MCM during GINS recruiting, the Sld3CBD-Cdc45 structure shows clearly that the Sld3CBD-binding site of Cdc45 is completely different from that of GINS and MCM binding to Cdc45. The model SCMG

confirmed such a binding manner, Sld3, Cdc45 and GINS could bind together. The competition of Sld3 and GINS for binding to Cdc45 or Cdc45-MCM reported by Bruck et. al, may be caused by the conformation change of Cdc45 DHHA1 between Sld3CBD-Cdc45 and CMG, or without other initiation factors (CMG formation is regulated by the initial factors). We modified the discussion (P10/L282-286). Regarding ssARS1-binding, we did not discuss with Bruck's data that ARS1-2 does not disrupt the Sld3-Cdc45 interaction, because the data does not conflict with our proposal, although the data does not have an advantage. We propose that the release of Sld3 and Sld7 from CMG could be associated with the binding of ssARS1 unwound by CMG, but the dissociation event of Sld3-Sld7 doesn't only ssARS1-binding. The exploration of unwound-ssARS1 causes the conformation change of CMG, which may be another event for Sld3-Sld7 dissociation. However, we do not have more experiments to confirm this and Bruck's ssDNA-binding experiment did not use all of Sld3, Cdc45 and MCM, so we do not discuss more with Bruck' data in the revised version (P11/L303-305),.

Reviewer #3 (Recommendations for the authors):

Major points:

(1) Figure 1, Sld3CBD-Cdc45 complex: Please indicate the number of critical residues and those of alpha-helices and beta-sheets in this Figure or Supplemental Figure to confirm the authors' claim.

Following the comment, we added the number of alpha-helices and beta-sheets with residue numbers in Figure 1, and Supplemental Figures 4 and 5. We also added a topology diagram (Supplemental Figure 3).

(2) Figure 2A and B: Please quantify the interaction here with a proper statistical comparison.

In the experiments of Figures 2A and 2B, we used a co-expression system to co-purify the complexes and check their binding. For quantifying, we added the concentrations of the samples used in the Method of [Mutant analysis of Sld3 and Cdc45].

(3) Figure 3B, EMSA: If these are from the EMSA assay, at least free DNAs and protein-bound DNAs are present on the gel. However, the authors showed one band, which seems to be free DNA in Figure 3B and separately the smear band of the protein complex in Supplementary Figure 12, and judged the DNA binding by the disappearance of the band (line 207). Interestingly, in the case of Sld3CBD, there are few smear bands (Supplementary Figure 12). Where is DNA in this case? The disappearance could be due to the contaminated nucleases (need a control non-specific DNA). Without showing the Sld3CBD-DNA complex in the gel, the conclusion that the DNA binding activity of Sld3CBD-Cdc45 to DNA is lower than Sld3CBD alone (line 210) is very much speculative. The same is true for Sld7-Sld3dC-Cdc45.

Please explain the method (EMSA) briefly in the main text and show a whole gel in both Figures. If the authors insist that the Sld3 DNA-binding activity is altered with Cdc43 (and MCM), it is better to perform a more quantitative DNA binding assay such as BIAcore (surface plasmon), etc.

In the EMSA, we use SYBR (Figure 4B) and CBB (Supplementary Figure 13) staining to show bands of ssDNA and protein, respectively. As the reviewer mentioned, the disappearance of the bands could be due to the contaminated nucleases, we did experiments with non-specific ssDNA-binding as a control using the same proteins shown in Supplementary Figure 14. So, we are convinced that the disappearance of the ssDNA bands or not disappearance could occur when binding to protein or not. We added such explanations in the text (P9/L242-244).

As we mentioned in the legend of Supplementary Figure 13, the Sld3CBD could not enter the gel, even when bound to ssDNA, because the pI values exceeded the pH of the running buffer.

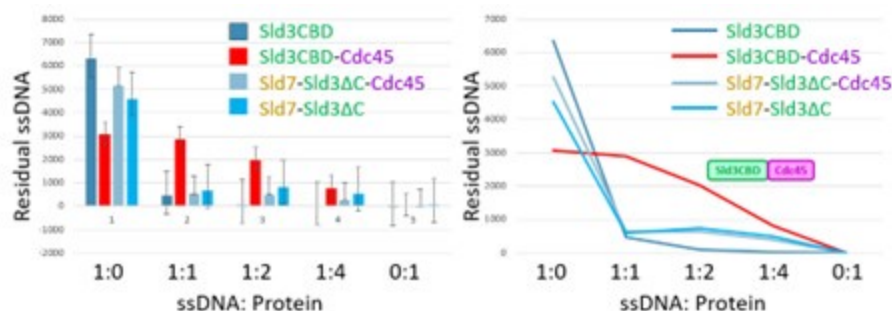
Following the reviewer's comments, we attempted a pull-down experiment using Histag (C-terminal histag of Sld3CBD/Sld3ΔC). Unfortunately, we encountered difficulties in achieving the balance between binding and chromatography conditions.

(4) Figure 3B: Please quantify the DNA binding here with a proper statistical comparison with triplicate.

For EMSA (Figure 3B), we used samples of ssDNA:protein= 1:0, 1:1, 1:2, 1:4 and 0:1 molecular ratios with 10 pM as a 1 unit. We added concentrations of the samples in the Method of [Electrophoretic mobility shift assay for ssDNA binding].

Following the comment, we tried to quantify the binding strength by integrating the grayscale of the bands in gel photos. However, we are concerned because this quantitative calculation through grayscale could not provide an accurate representation of results. Many sample groups cannot be run on one gel. Therefore, the gel differences in parameters cause large errors in the calculation as shown in Author response image 1. Although the calculated integral grayscale chart is consistent with our conclusion, we do not want to add this to our manuscript.

Author response image 1.



(5) Because of poor writing, the authors need to ask for English editing.

We are very sorry for the language. We asked a company (Editag, <https://www.editage.jp>) to do a native speaker revision and used AI to recheck English.

Minor points:

(1) Lines 47-58, Supplementary Figure 1: Although the sentences describe well how CMG assembles on the replication origin, the figure does not reflect what is written, but rather shows a simple schematic figure related to the work. However, for the general readers, it is very useful to see a general model of the CMG assembly. Then, the authors need to emphasize the steps focused in this study.

Thank you for your thoughtful comments. We optimized Figure 1 and hope it will be more understandable to general readers.

(2) Line 50, DDK[6F0L](superscript): what is 5F0L?

We are sorry for this mistake, that is a PDBID of the DDK structure. we deleted 6F0L.

(3) Lines 68 and 69, ssDNA and dsDNA: should be "single-stranded DNA (ssDNA)" and double-stranded DNA (dsDNA) when these words appear for the first time.

Following the comment, we modified it to "single-stranded DNA (ssDNA)" and "double-stranded DNA (dsDNA)" (P3/L68,70).

(4) Line 84, Cdc45s: What "s" means here?

We are sorry for this mistake, we modified it to "Cdc45".

(5) Line 87, Sld3deltaC: What is Sld3deltaC? This is the deletion of either the Cdc45-binding domain or the C-terminal domain.

Sld3ΔC is a deletion of the C-terminal domain of Sld3. We added the residue range and explanation (P4/L91).

(6) Line 103: Although the authors mentioned beta-sheets 1-14 in the text, there is no indication in Figures. It is impossible to see the authors' conclusion.

The secondary structure elements of Sld3CBD-Cdc45 are shown in Supplementary Figures 4 and 5. Following the comment, we added a topology diagram of Sld3CBD and Cdc45 in the Sld3CBD-Cdc45 complex as Supplementary Figure 3 and added citations when describing structural elements.

(7) Line 106, huCdc45: Does this mean human Cdc45? If so, it should be "human CDC45 (huCDC45). CMG form is from budding yeast? Please specify the species.

Yes, huCdc45 is human Cdc45. We modified it into "human CDC45 (huCdc45)".

(8) Line 107, Supplemental Figure 3B, black ovals: Please add "alpha7" in the Figure.

Following the comment, we added a label of Cdc45 α7 to Supplemental Figure 3B and 3C (Supplemental Figure 4B and 4C in revised version).

(9) Line 128, DHHA1: What is this? Please explain it in the text.

Following the comment, we added the information on DHHA1 (P3/L75-77).

(10) Line 130, beta13, and beta14: If the authors would like to point out these structures, please indicate where these sheets are in Figures.

We added a topology diagram as Supplementary Figure 3 to show the β-sheet in DHH and added a citation in the text.

(11) Line 133: Please add (Figure 1B) after the α8CTP.

Following the comment, we added "(Figure 1C)" (1B is 1C in revised version) after the α8CTP (P6/L133).

(12) Line 140: After DHHA1, please add (Figure 1C).

Following the comment, we added the figure citation after the DHHA1 (P6/L140).

(13) Line 142: After DHHA1, please add (Figure 1D).

Following the comment, we added the figure citation after the DHHA1 (P6/L142).

(14) Line 149, *Sld3-Y seemed to retain a faint interaction with Cdc45. The Cdc45 band is too faint here. Moreover, as shown above, without the quantification with proper statistics, it is hard to draw this kind of conclusion.*

We agree that the Cdc45 band corresponding to Sld3-Y in the pull-down assay was very faint, so we performed an *in vivo* experiment (Fig2C) to confirm this result.

(15) Line 149, Figure 2A and B: *What kind of interaction assay was used here? Simple pull-down. It seems to eluate from the column. If so, how do the authors evaluate the presence of the proteins in different fractions? Please explain the method briefly in the main text.*

Figure 2 shows a co-express pull-down binding assay. To describe the co-express pull-down experiments clearly, we added more explanations in the Methods [Mutation analysis of Sld3 and Cdc45].

(16) Line 154-155: *Please show the quantification to see if the reduced binding is statistically significant.*

Here, we explain why Cdc45-A remained Sld3CBD-bind ability. Although mutant Cdc45-A has reduced three hydrogen bonds with D344 of Sld3CBD, the remaining hydrogen-bond network keeps contact between Sld3CBD and Cdc45.

(17) Line 158, cell death: *"No growth" does not mean cell death. Please rephrase here.*

Following the comment, we modified it to "no growth" (P6/L158).

(18) Line 166: After CMG dimer, please add "respectively".

Following the comment, we added the word " , respectively" after CMG dimer (P7/L178).

(19) Line 194-195: *I can not catch the meaning. Please rephrase here to clarify the claim. What are ssARS1-2 and ARS1-5?*

Following the comment, we added more information about ssDNA fragments at the beginning of this section (P8/L210-214).

(20) Figure 4A and Supplemental Figure 12 top, schematic figure of ARS region. *It is hard to catch. More explanation of the nature of the DNA substrates and much better schematic presentations would be appreciated.*

Following the comment, we added more information about ARS1 to the figure legend.

(21) Figure 1A, dotted ovals should be dotted squares as shown in the enlarged images on the bottom.

Following the comment, we modified Figure 1A and the legend to change the dotted ovals into dotted squares.

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